KU LEUVEN

Chapter 2.
Eploratory Data Analysis & Preprocessing

Data Science for Business

D0I74A

Prof. dr. Wouter Verbeke

Prof. dr. Bart Baesens



Course schedule

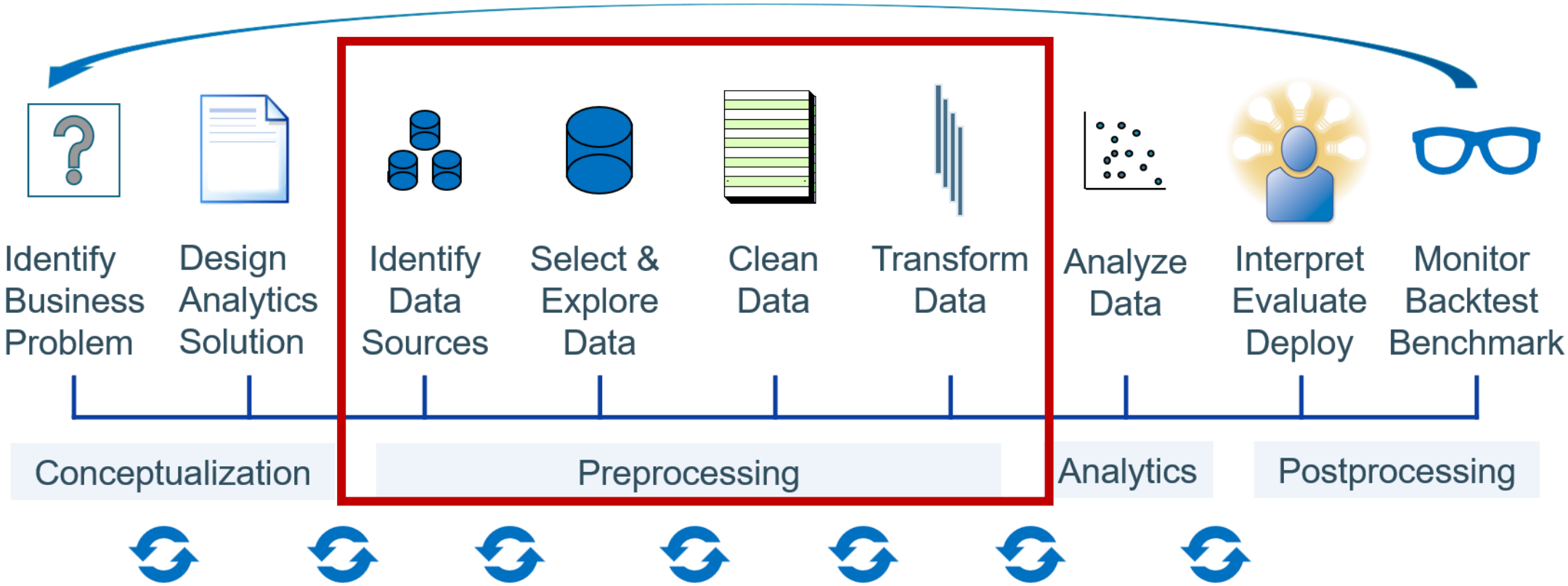
BLOCK1

1	21/09/2026	Chapter 0. Course Outline Chapter 1. Data Science for Business	
1	22/09/2026	Chapter 1. Data Science for Business (ctd.)	
		Hands-on #1: Building a first pipeline	**Homework 1**
2	28/09/2026	Chapter 1. Data Science for Business (ctd.) Chapter 2. Exploratory Data Analysis & Preprocessing	
2	29/09/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
		Hands-on #2: EDA & Preprocessing	**Homework 2**
3	05/10/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.) Chapter 3. Predictive Analytics - Decision Trees	
3	06/10/2026	Chapter 4. Predictive Analytics - Nearest Neighbors & Ensembles	
		Hands-on #3: Predictive models	**Homework 3**
4	12/10/2026	Chapter 5. Postprocessing	
4	13/10/2026	Chapter 5. Postprocessing	
		Hands-on #4: Postprocessing	**Homework 4**

Data science process model

Chapter 2

- Practical steps in elaborating a customer churn prediction model:



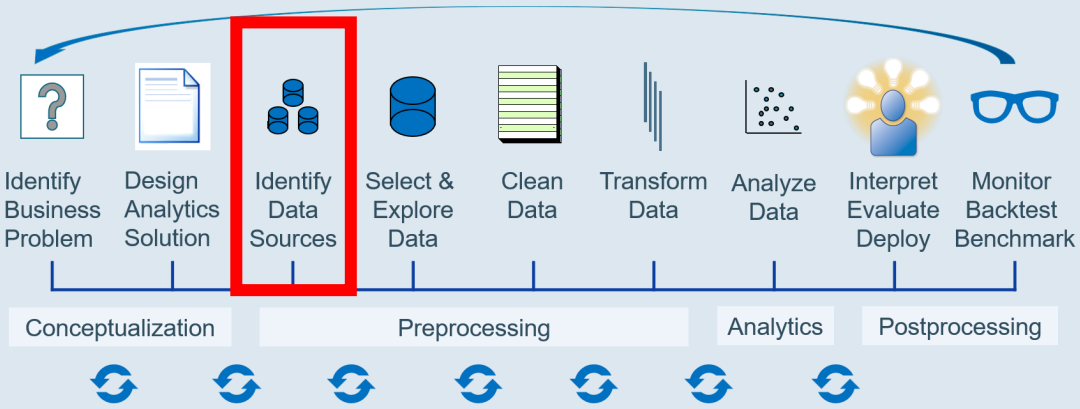
Outline

1. Identify data
2. Select data
3. Explore data
4. Select data
5. Clean data
6. Transform data

Learning objectives

At the end of this chapter, students should be able to:

- Understand and explain the need for preprocessing data
- Understand and explain the impact of various preprocessing techniques on the resulting analytical model
- Understand and explain the importance of sampling (bias)
- Understand and explain the purpose and workings of the discussed preprocessing techniques, in particular, the weights-of-evidence approach and the associated information value



1. Identify data

Identify data

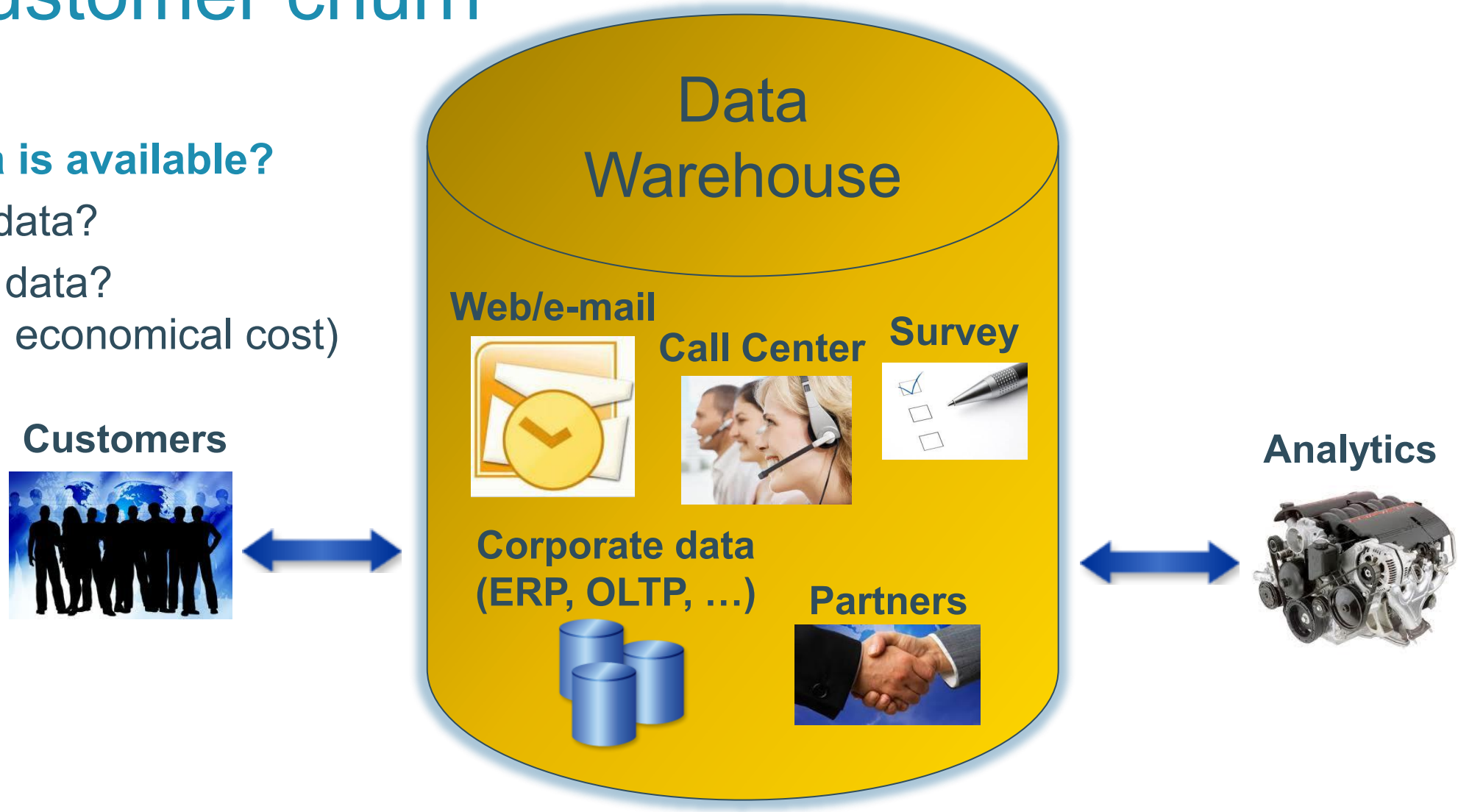


- **Technical knowledge** required: **what data is available?**
  - Information system architecture:
    - Both **applications** (sources) and **databases** (storage)
    - Involvement of database administrators required!
  - Analytical solutions
    - What data could be processed by which analytical methods?

- **Business knowledge** required: **what data could be useful?**
  - Understanding of the business problem
  - What **information** would be useful towards improving decision-making?

Case: customer churn

1. **What data is available?**
   - Internal data?
   - External data?
     (cf. infra, economical cost)

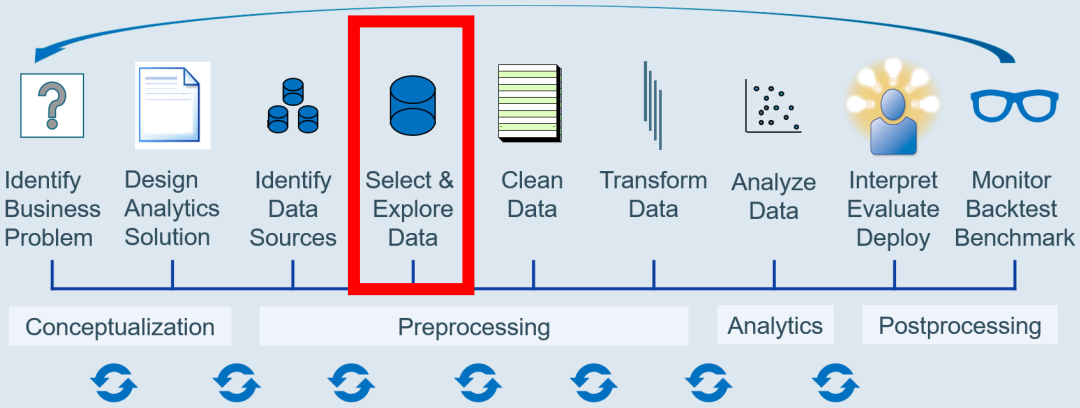


Case: customer churn

2. What data could be useful?

- Use **business expertise** to uncover potentially relevant predictors: talk to experts!
- The **more**, the better
- The **more diverse**, the better
  - Cover different *factors* that ***explain*** and allow to ***predict*** customer behavior
  - **Guideline: diversity!**
- Case: customer churn (telco benchmarking study, Verbeke et al., 2012)
  - Usage
  - Financial
  - Marketing
  - Socio-demographic
  - Social network

Verbeke, W., Dejaeger, K., Martens, D., Hur, J., & Baesens, B. (2012). New insights into churn prediction in the telecommunication sector: A profit driven data mining approach. *European journal of operational research*, *218*(1), 211-229.



2. Select data

Terminology

Flattening data

Sampling

Select data

- Objective: **structured dataset**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	State	A_Length	Area_Code2	Intl_Plan	Vmail	Vmail_	Day_Mins	Day_Calls	Day_Charge	Night_Mins	Night_Calls	Night_Charge	Intl_Mins	Intl_Calls	Intl_Charge	CustServ	Churn
2	KS	128	AC415	no	yes	25	265.1	110	45.07	244.7	91	11.01	10	3	2.7	1	False.
3	OH	107	AC415	no	yes	26	161.6	123	27.47	254.4	103	11.45	13.7	3	3.7	1	False.
4	NJ	137	AC415	no	no	0	243.4	114	41.38	162.6	104	7.32	12.2	5	3.29	0	False.
5	OH	84	AC408	yes	no	0	299.4	71	50.9	196.9	89	8.86	6.6	7	1.78	2	False.
6	OK	75	AC415	yes	no	0	166.7	113	28.34	186.9	121	8.41	10.1	3	2.73	3	False.
7	AL	118	AC510	yes	no	0	223.4	98	37.98	203.9	118	9.18	6.3	6	1.7	0	False.
8	MA	121	AC510	no	yes	24	218.2	88	37.09	212.6	118	9.57	7.5	7	2.03	3	False.
9	MO	147	AC415	yes	no	0	157	79	26.69	211.8	96	9.53	7.1	6	1.92	0	False.
10	LA	117	AC408	no	no	0	184.5	97	31.37	215.8	90	9.71	8.7	4	2.35	1	False.
11	WV	141	AC415	yes	yes	37	258.6	84	43.96	326.4	97	14.69	11.2	5	3.02	0	False.
12	IN	65	AC415	no	no	0	129.1	137	21.95	208.8	111	9.4	12.7	6	3.43	4	True.
13	RI	74	AC415	no	no	0	187.7	127	31.91	196	94	8.82	9.1	5	2.46	0	False.
14	IA	168	AC408	no	no	0	128.8	96	21.9	141.1	128	6.35	11.2	2	3.02	1	False.
15	MT	95	AC510	no	no	0	156.6	88	26.62	192.3	115	8.65	12.3	5	3.32	3	False.
16	IA	62	AC415	no	no	0	120.7	70	20.52	203	99	9.14	13.1	6	3.54	4	False.
17	NY	161	AC415	no	no	0	332.9	67	56.59	160.6	128	7.23	5.4	9	1.46	4	True.
18	ID	85	AC408	no	yes	27	196.4	139	33.39	89.3	75	4.02	13.8	4	3.73	1	False.
19	VT	93	AC510	no	no	0	190.7	114	32.42	129.6	121	5.83	8.1	3	2.19	3	False.
20	VA	76	AC510	no	yes	33	189.7	66	32.25	165.7	108	7.46	10	5	2.7	1	False.
21	TX	73	AC415	no	no	0	224.4	90	38.15	192.8	74	8.68	13	2	3.51	1	False.
22	FL	147	AC415	no	no	0	155.1	117	26.37	208.8	133	9.4	10.6	4	2.86	0	False.
23	CO	77	AC408	no	no	0	62.4	89	10.61	209.6	64	9.43	5.7	6	1.54	5	True.
24	AZ	130	AC415	no	no	0	183	112	31.11	181.8	78	8.18	9.5	19	2.57	0	False.
25	SC	111	AC415	no	no	0	110.4	103	18.77	189.6	105	8.53	7.7	6	2.08	2	False.
26	VA	132	AC510	no	no	0	81.1	86	13.79	237	115	10.67	10.3	2	2.78	0	False.
27	NE	174	AC415	no	no	0	124.3	76	21.13	250.7	115	11.28	15.5	5	4.19	3	False.
28	WY	57	AC408	no	yes	39	213	115	36.21	182.7	115	8.22	9.5	3	2.57	0	False.
29	MT	54	AC408	no	no	0	134.3	73	22.83	102.1	68	4.59	14.7	4	3.97	3	False.

Select data: terminology

Target variable, a.k.a., (predictive analytics)
- Outcome
- Output
- Dependent variable (statistics)
- Class variable (if categorical)

Observation of a **unit** or **entity**, a.k.a.,
- Record (SQL)
- Row
- Line
- Instance
- Data point (cfr. scatter plot)
- Sample
- …

Variable, a.k.a.,
- Field (SQL)
- Column
- Input
- Independent variable (stat.)
- Predictor (pred. analytics)
- Attribute
- Characteristic
- Feature
- Factor
- …

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Select data: flattening data

Cf. Target definition & feature engineering:
- **Handling the time-dimension in the data**
- **Time-flattening the data!**

- Merging data into a ***flat table***, i.e., a 2-dimensional (= flat) ***tabular data set***
  - E.g., starting from a relational data-base:

Transactions		
ID	Date	Amount
XWV	2/01/2015	52 €
XWV	6/02/2015	21 €
XWV	3/03/2015	13 €
BBC	17/02/2015	45 €
BBC	1/03/2015	75 €
VVQ	2/03/2015	56 €

Customer data		
ID	Age	Start date
XWV	31	1/01/2015
BBC	49	10/02/2015
VVQ	21	15/02/2015

Non-normalized data table				
ID	Date	Amount	Age	Start date
XWV	2/01/2015	52 €	31	1/01/2015
XWV	6/02/2015	21 €	31	1/01/2015
XWV	3/03/2015	13 €	31	1/01/2015
BBC	17/02/2015	45 €	49	10/02/2015
BBC	1/03/2015	75 €	49	10/02/2015
VVQ	2/03/2015	56 €	21	15/02/2015

Unit of observation: transaction!

Choice! → **design the solution**!

Select data: sampling

- Sampling = selecting observations
- Sample = set of observations that will be further analyzed
  e.g., to develop a customer churn prediction model
  (note: sometimes, ‘a sample’ refers to a single observation)

- Selection to be performed across **application-dependent dimensions**:
  - Customer segments
  - Channels
  - Time (cf. infra)
  - …

  → **Design the solution!**

Select data: sampling

Question: is it possible that the sample = population?

- **Key guideline for selecting a sample**:

The sample should be REPRESENTATIVE

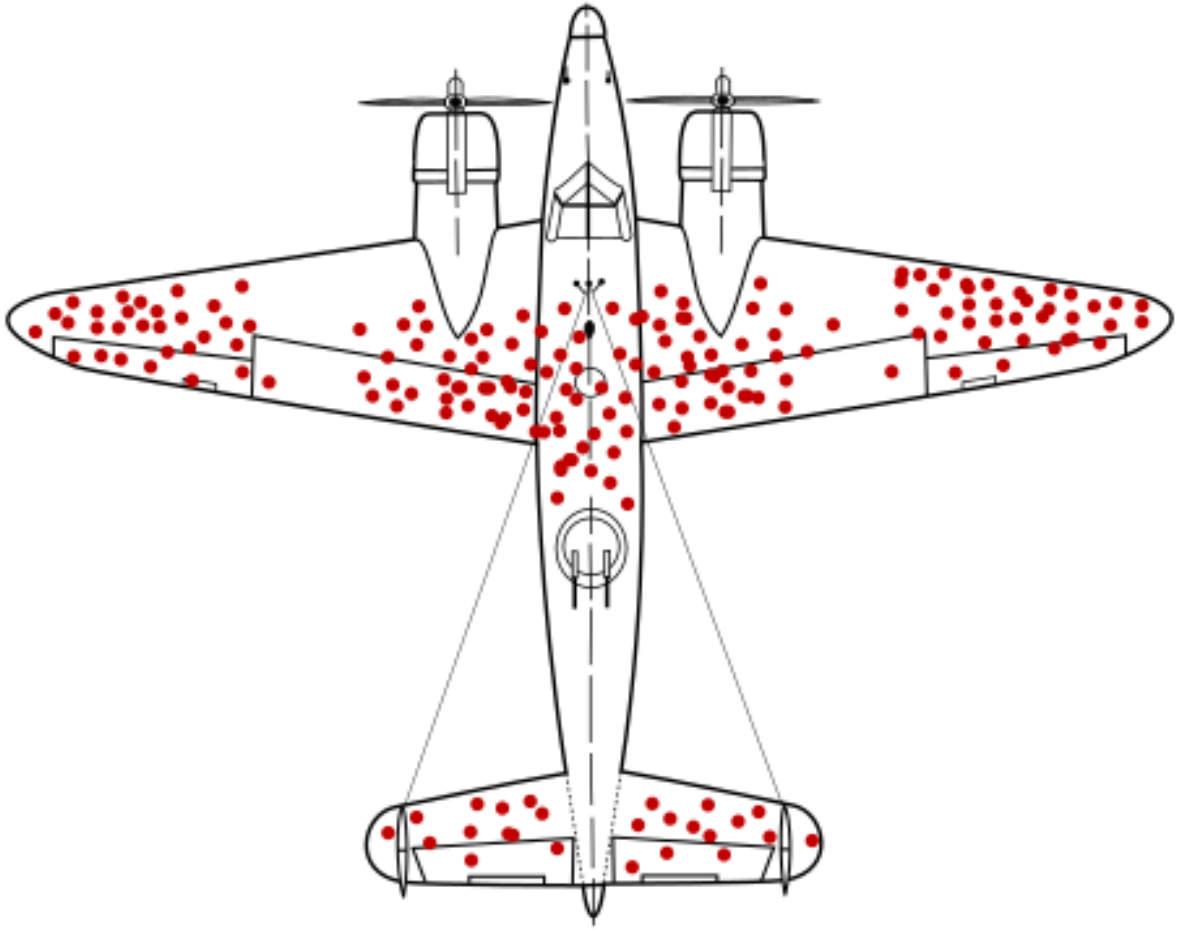
OF THE POPULATION

ON WHICH THE MODEL WILL BE APPLIED

- Not all samples are representative, then?
  - Beware: sources of bias!
  - Think of the ***data generating process (DGP)*** to identify **sources of bias**

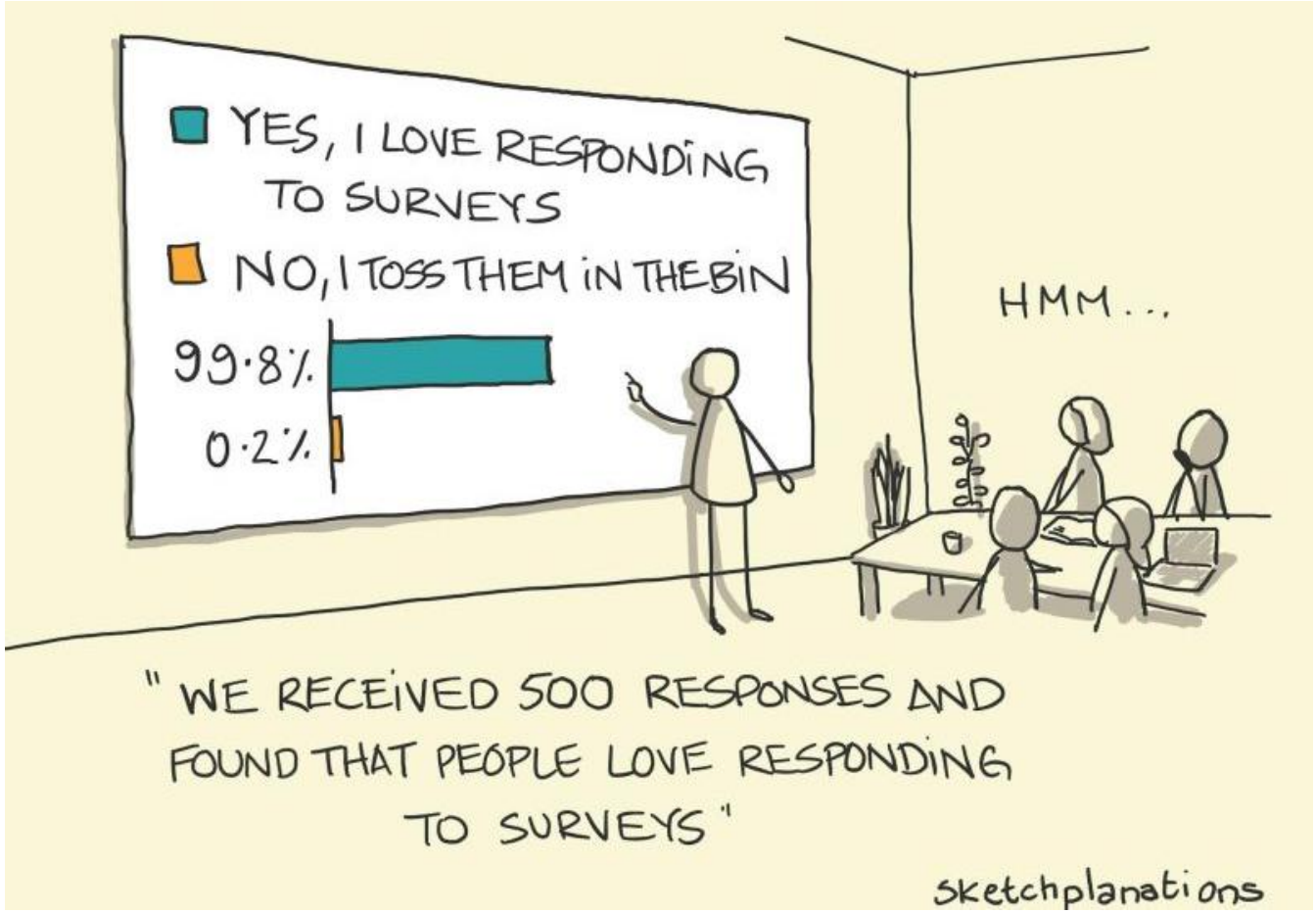
Select data: sampling

Survivorship bias



https://peopleplusscience.com/the-missing-bullet-holes-how-a-wwii-statisticians-insight-can-save-your-strategy/
https://en.wikipedia.org/wiki/Survivorship_bias

Select data: sampling

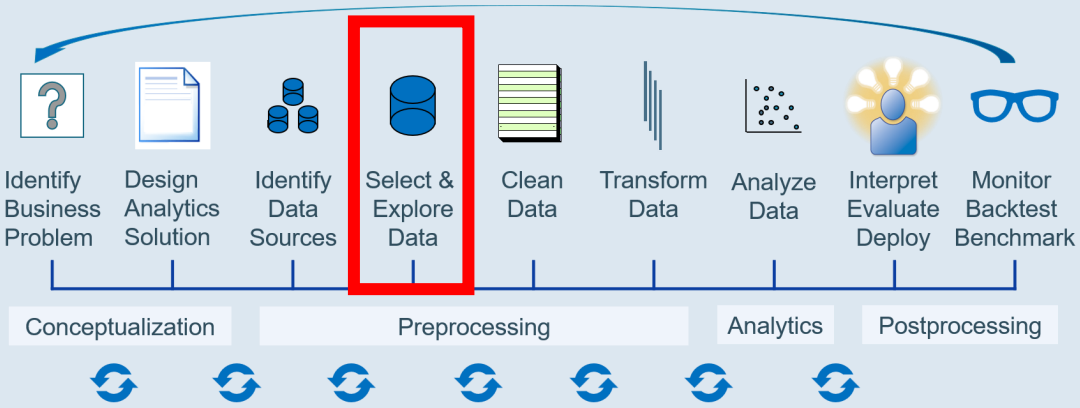


https://peopleplusscience.com/the-missing-bullet-holes-how-a-wwii-statisticians-insight-can-save-your-strategy/
https://en.wikipedia.org/wiki/Survivorship_bias

Select data: sampling

- Bias?
  - Credit card transactional fraud detection
  - Application credit scorecard development

- Types of bias:
  - **Seasonality effects**: consider the handling of time
  - Sample **selection bias**: consider the selection mechanism



3. Explore data

Data understanding

Plots

Data understanding

What do we mean with *optimal*?

Understanding the data is paramount for **optimal preprocessing**

- For gaining understanding:
  - **Data dictionary**
    - **Metadata**: information on variables (definition, measurement scale, …)
  - **Data exploration**
    - Distributions
    - Relations
    - …
- To guide **decisions** in preprocessing

Data understanding: measurement scale

Scale	Meaning	Example
Nominal	“Name” of objects	Cities
Ordinal	“Ordering” objects (no distance)	Range of income XS-S-M-L-XL
Interval	Zero and unity arbitrary	Temperature in grades Celsius
Proportional	Zero given and unity arbitrary	Weight in kg Income in $
Absolute	Zero and unity given	Counting objects

Data understanding: measurement scale

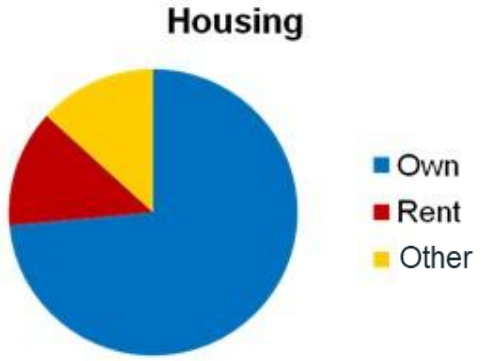
Scale	
Nominal	Categorical variables
Ordinal	
Interval	Continuous variables
Proportional	
Absolute	

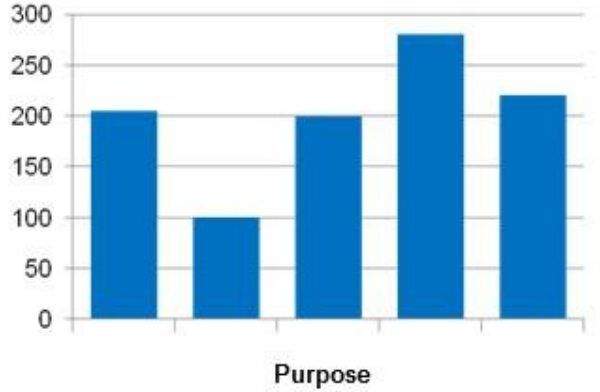
Why is this important?

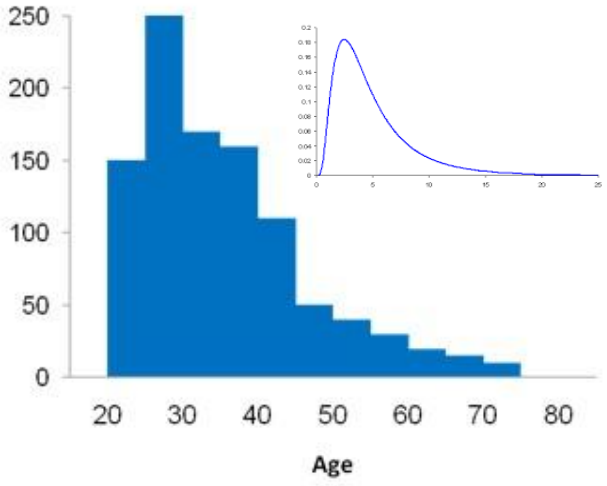
- Some techniques **cannot handle categorical variables**
  - Then categorical variables have to be made continuous, i.e., transformed
  - **Note**:
    - Categorical variable with two possible values is a **binary categorical variable**, which …
    - Can be easily zero-one encoded so as to get a **dummy variable**, which …
    - Can be treated as if a **continuous variable**
- Some techniques **cannot handle continuous variables**
  - Then continuous variables have to be **binned**, i.e., made categorical (a.k.a. categorization, segmentation, …)

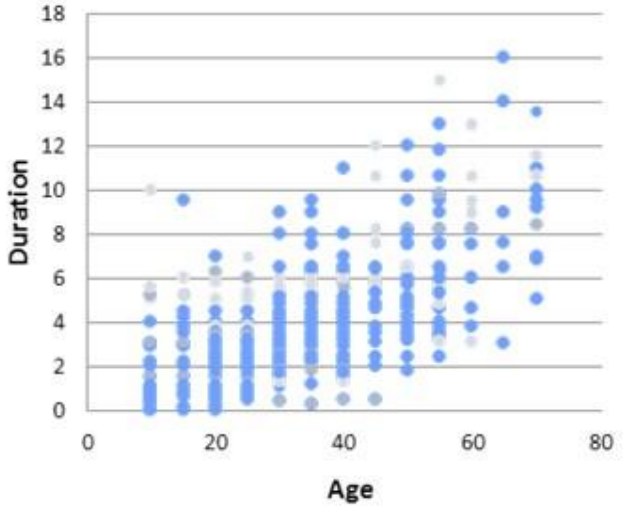
Plots

- Cf. SEMMA
  - E: exploration
- Visual analytics
  - Tableau, SAS, …
- Distributions
  - Pie charts, histograms, …
- Relations
  - Correlation plots, …









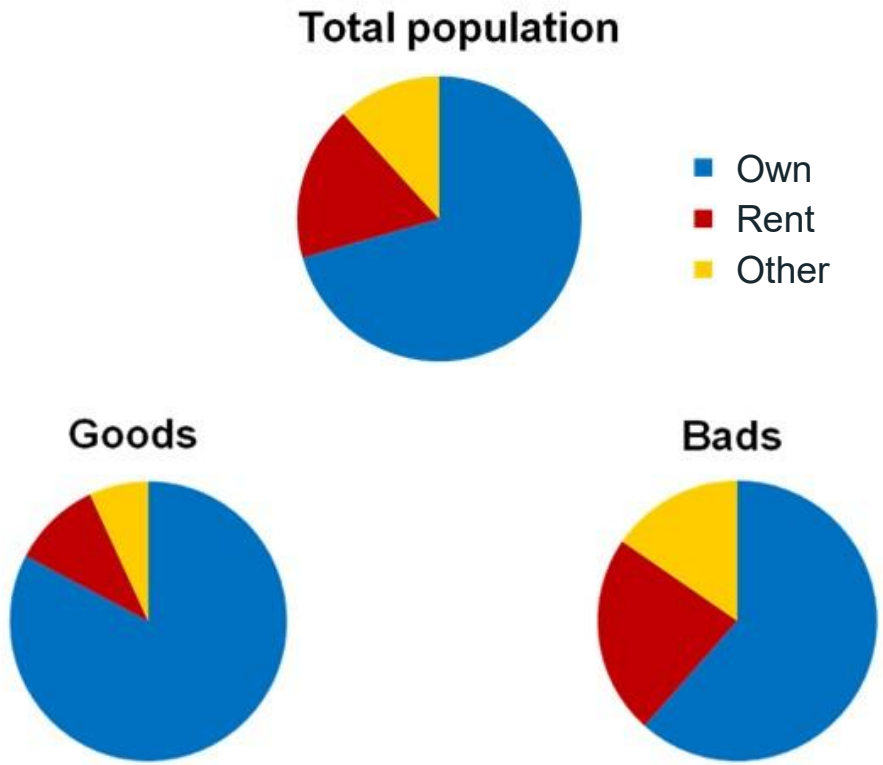
https://www.tableau.com/

https://www.sas.com/en_us/software/visual-analytics.html

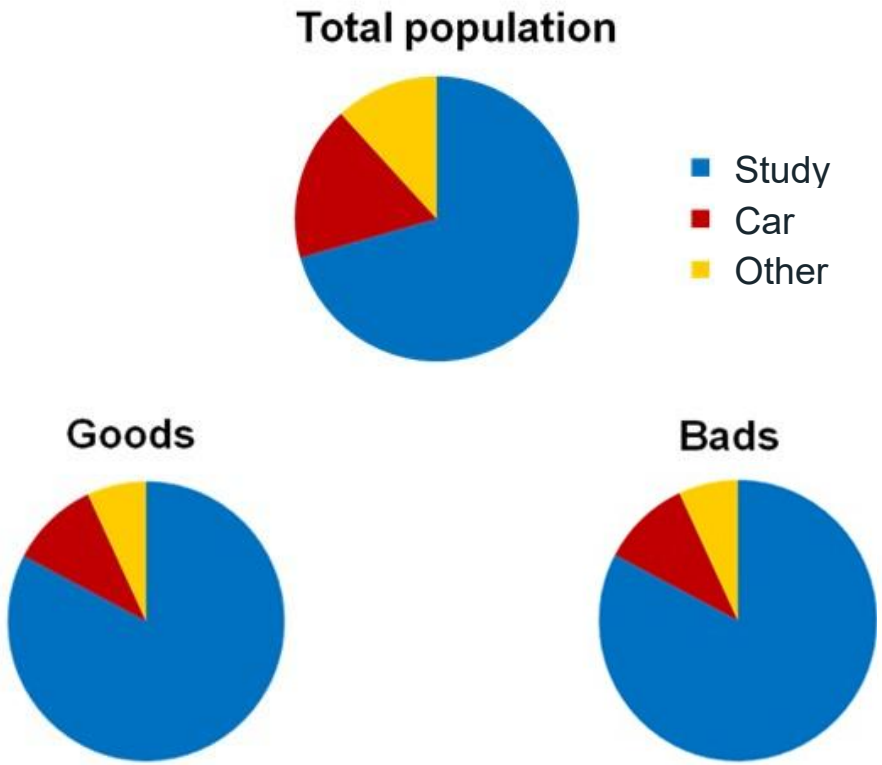
Plots: distributions

What do you see?
Why is that interesting to know?

Housing

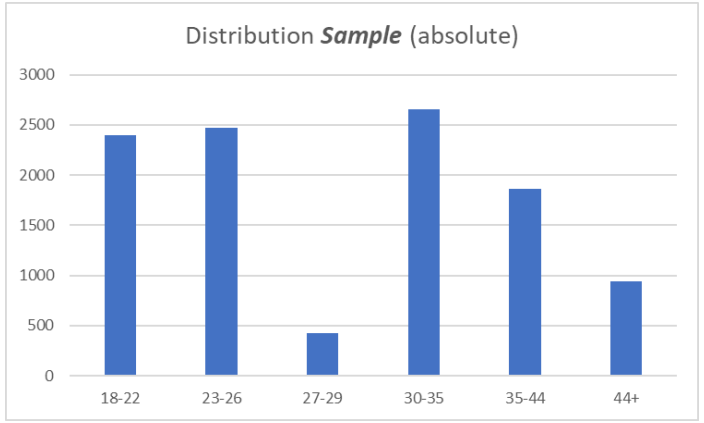


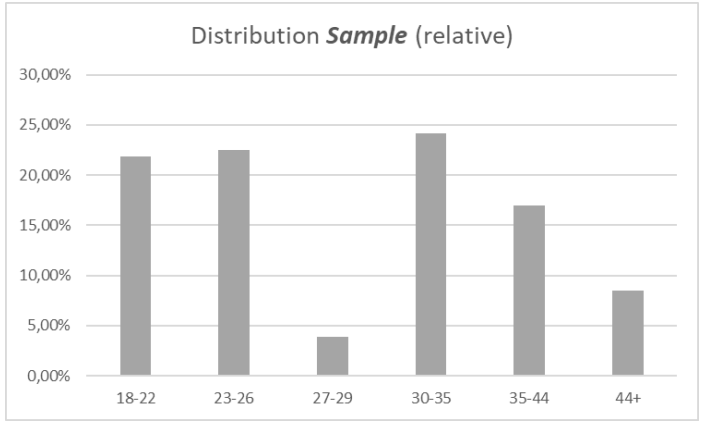
Purpose



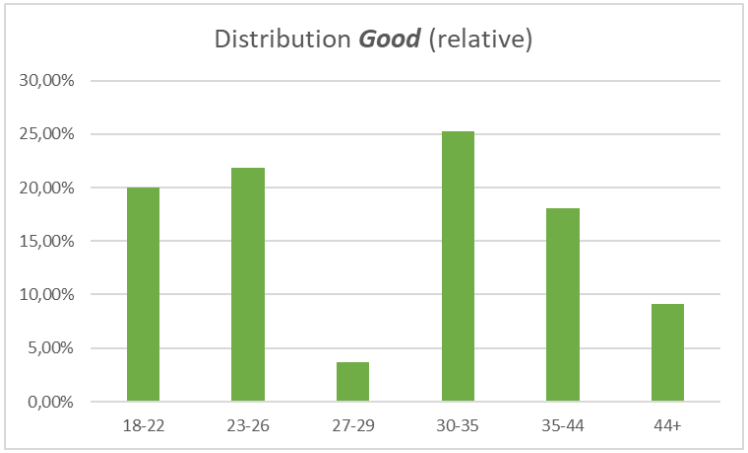
Plots: distributions

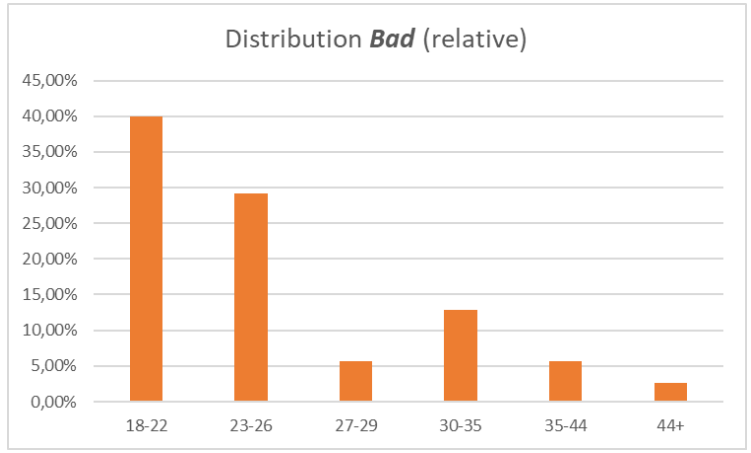
Age	Count	Distr. Count	Good	Distr. Goods	Bad	Distr. Bads
Missing	240	2,18%	200	2,00%	40	4,00%
18-22	2400	21,82%	2000	20,00%	400	40,00%
23-26	2475	22,50%	2183	21,83%	292	29,20%
27-29	424	3,85%	368	3,68%	56	5,60%
30-35	2659	24,17%	2530	25,30%	129	12,90%
35-44	1864	16,95%	1807	18,07%	57	5,70%
44+	938	8,53%	912	9,12%	26	2,60%
Total:	**11000**		**10000**		**1000**	





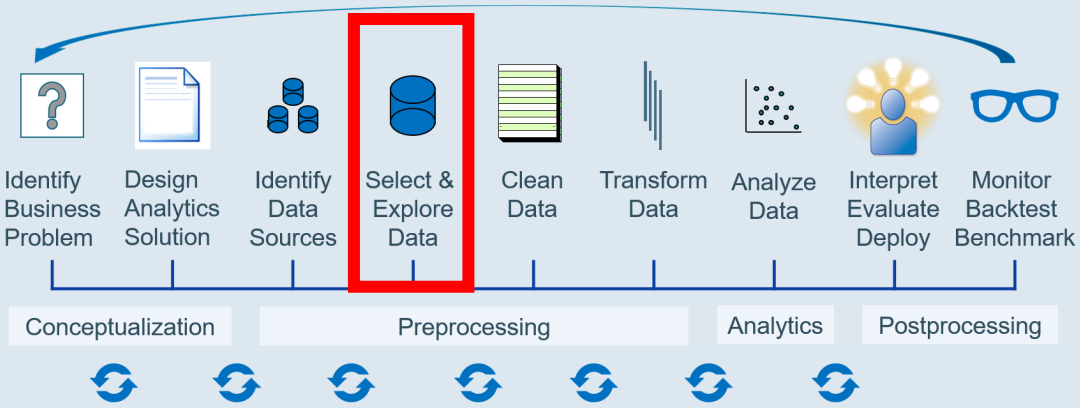
What do you see? Why is that interesting to know?





Note: distributions in relative numbers (%) are easier to compare, since they have the same scale you may get to insights

KU LEUVEN



4. Select data

Variable selection

Target variable

Select data: variable selection

- SQL: records and fields
  - → record = observation
  - → field = variable
- Sampling = selecting records
- Variable selection = selecting fields

Non-normalized data table				
ID	Date	Amount	Age	Start date
XWV	2/01/2015	52 €	31	1/01/2015
XWV	6/02/2015	21 €	31	1/01/2015
XWV	3/03/2015	13 €	31	1/01/2015
BBC	17/02/2015	45 €	49	10/02/2015
BBC	1/03/2015	75 €	49	10/02/2015
VVQ	2/03/2015	56 €	21	15/02/2015

- The more variables in the data set, the better …
  - → At least, to start with!
  - → Depending on the method, **variable selection** may need to be applied

Select data: variable selection

- Depending on the method, **variable selection** may need to be applied
  - Some methods have **built-in variable selection procedures**
    - E.g., Decision Trees (cf. Chapter 3)
  - Some methods do not
    - E.g., Artificial Neural Networks (cf. Chapter 6)
  - These may ***benefit*** from variable selection
    - I.e., from reducing the number of variables in the model
  - ***Benefit?***
    - Predictive power
    - Interpretability (depending on the method)
    - Stability (cf. infra)
    - …

Select data: variable selection

- Why may predictive power improve because of variable selection?
  - '***Noise***' in the dataset: irrelevant variables in dataset
    - Including irrelevant variable in a model → irrelevant information used to predict
    - Obscures true signals
    - Make it harder for learners to distill *true* relationships from data, to *fit the model well*
  - **Curse of dimensionality**
    - Number of training observations required to fit a model
    - Increases exponentially with the number of variables
    - Including many variables in model requires a lot of data
      - If not enough data, then fitted coefficients and model will be *unstable*, have high variance
  - **Reducing multicollinearity** yields lower variance (cf. infra, stability)

Select data: variable selection

50 variables
→ 1.125.899.906.842.623 combinations

What is a **heuristic?**

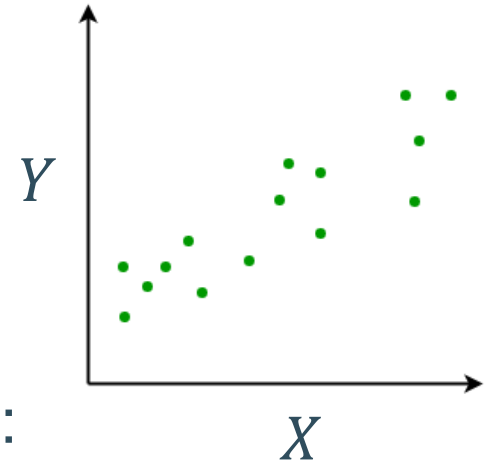
- Objective in **variable selection**: finding the **optimal set of variables**
  - If n **variables** are present, $\mathbf{2^n - 1}$ **possible variable sets** can be considered
    - Every variable can be in or out the model
    - A binary sequence can represent whether variables are in or out
      - E.g., with 5 variables: 0 1 0 1 1 meaning variables 1 and 3 are out, variables 2, 4 and 5 are in
    - With n variables, this leads to 2^n possible sequences, combinations or variable sets
    - The sequence with all zeros is not a valid possibility, then no variables are included
    - Hence, there are $2^n - 1$ possible variable sets
  - **Heuristic search methods** are needed!
  - Good variable subsets:
    - Contain variables highly correlated with (i.e., are predictive of) the class
    - That are uncorrelated with (not predictive of) each other (why? cf. infra: stability)

Select data: variable selection

- **Filters** to reduce the # of variables in the dataset (and as such, in the model)
  - Filter using a measure of the strength of association between two variables
  - Prior to fitting a model → **preprocessing step**

	Continuous target	Discrete target
Continuous variable	• Pearson correlation	• Fisher score
Categorical variable	• Fisher score • ANOVA analysis (cf. Statistics course)	• Information value (cf. infra) • Gain/entropy (cf. infra)

Select data: variable selection



- **Pearson correlation**
  - Pearson correlation between variables X and Y is defined as:

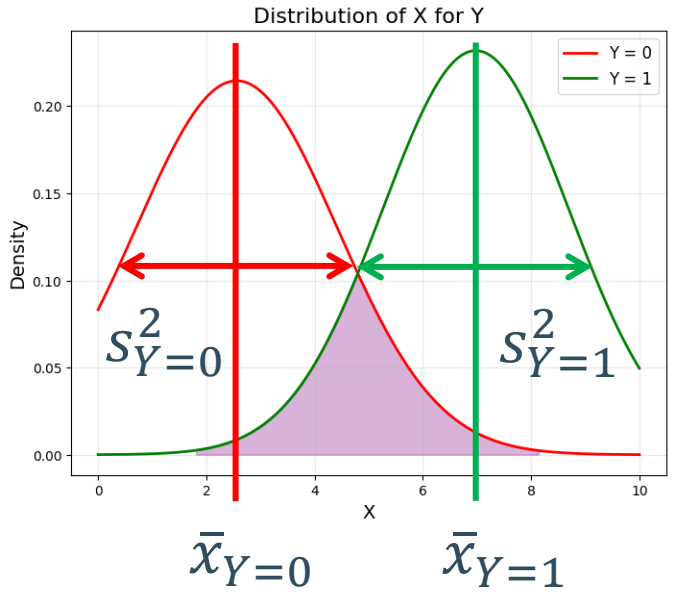
$$\rho_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

  - Evaluates strength of **linear relation** between **two continuous variables**
    - E.g., between continuous predictor (X) and continuous target (Y)
    - Always varies between -1 and +1
  - Use for filtering:
    - Keep variables with |ρ| > 0.50
    - Keep top 10% of variables with strongest correlation

Select data: variable selection

- **Fisher score**
  - Defined as:

$$Fisher\ score = \frac{|\bar{x}_{Y=0} - \bar{x}_{Y=1}|}{\sqrt{s^2_{Y=0} + s^2_{Y=1}}}$$

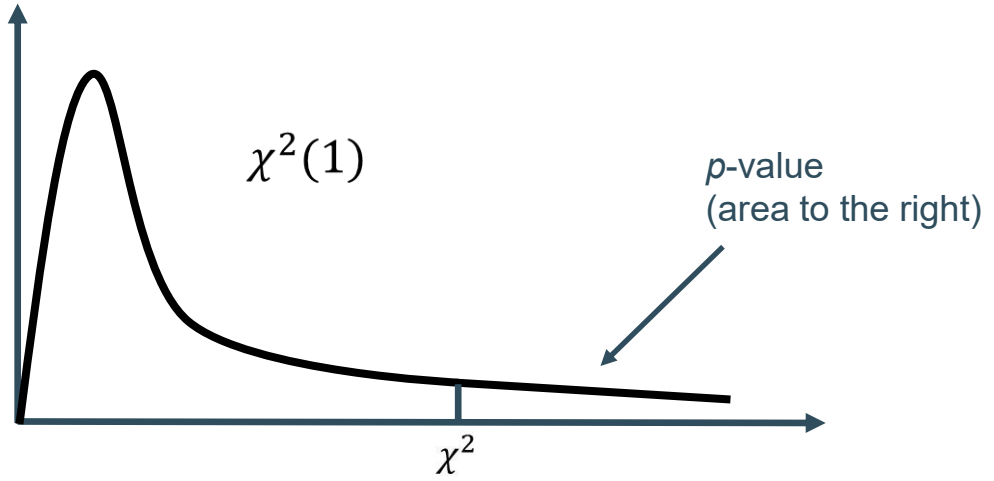


  - A higher score indicates a more predictive variable
  - Can also be used if the target is continuous and variable is discrete

Select data: variable selection

- **Forward/Backward/Stepwise Logistic Regression**
  - Perform a hypothesis test (Wald test) to decide upon variable importance:
    - H_0: $\beta_i = 0$
    - H_1: $\beta_i \neq 0$
  - Under H_0, the following can be shown:

$$\left(\frac{\widehat{\beta_i}}{s.e.(\widehat{\beta_i})}\right)^2 \sim \chi^2(1)$$



Select data: variable selection

- **Forward/Backward/Stepwise Logistic Regression**
  - Small *p*-value means important variable
  - Large *p*-value means unimportant variable
  - Use the *p*-value to decide on the importance of the variables:
    - *p*-value < 0.01: highly significant
    - 0.01 < *p*-value < 0.05: significant
    - 0.05 < *p*-value < 0.10: weakly significant
    - 0.1 < *p*-value: not significant

Select data: variable selection

- **Forward/Backward/Stepwise Logistic Regression**
  - Forward selection
    - Starts from empty model and always adds variables based on low p-values
  - Backward elimination
    - Starts from full model and always removes variables based on high p-values
  - Stepwise
    - Starts as forward selection, but checks whether added variables can be removed later

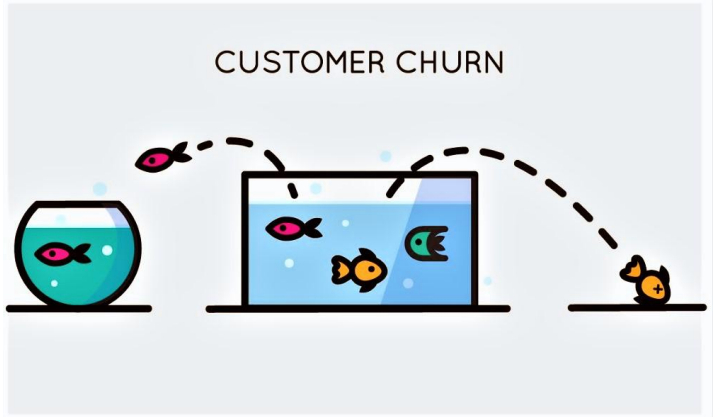
Select data: target variable

- Predictive analytics (supervised learning): **target variable (Y)**
- Observed outcome

  … or outcome ***to be defined***?

- Customer churn prediction:
  - Contractual vs. non-contractual setting

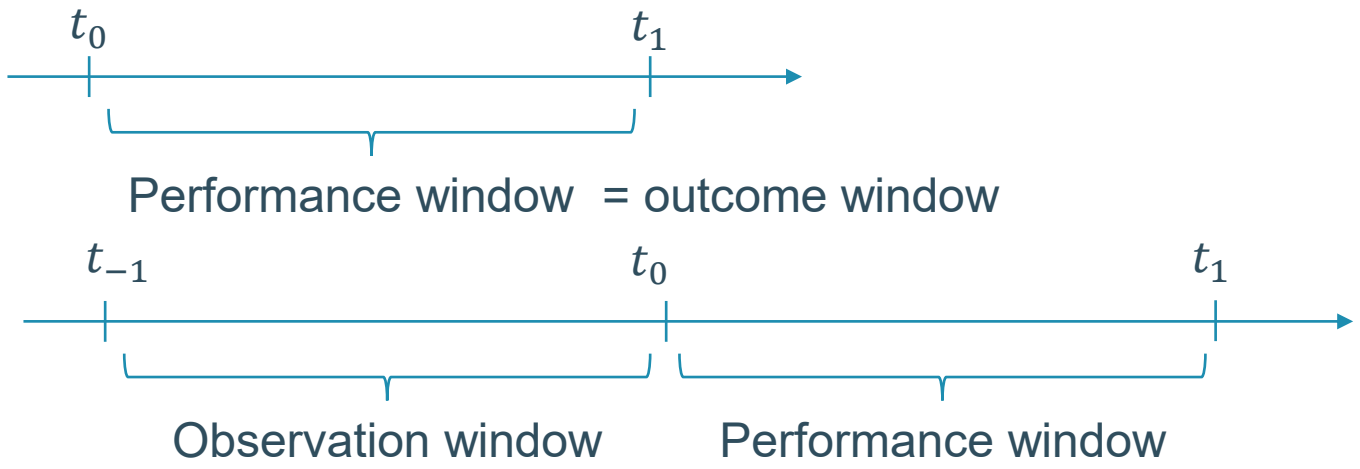
    → **“proxy” variable**



Case: customer churn

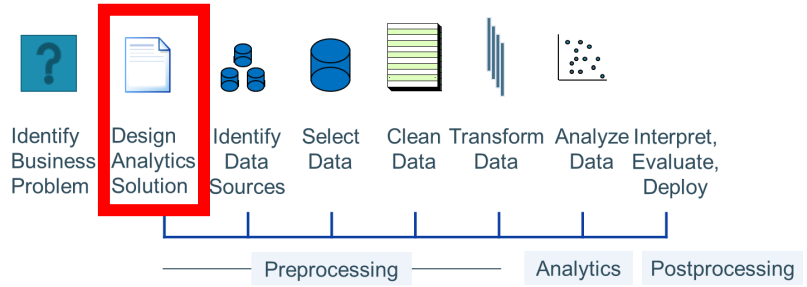
Performance window: $t_1 - t_0 = ?$
Observation window : $t_0 - t_{-1} = ?$

- Definition target variable: how to handle the **time dimension?**



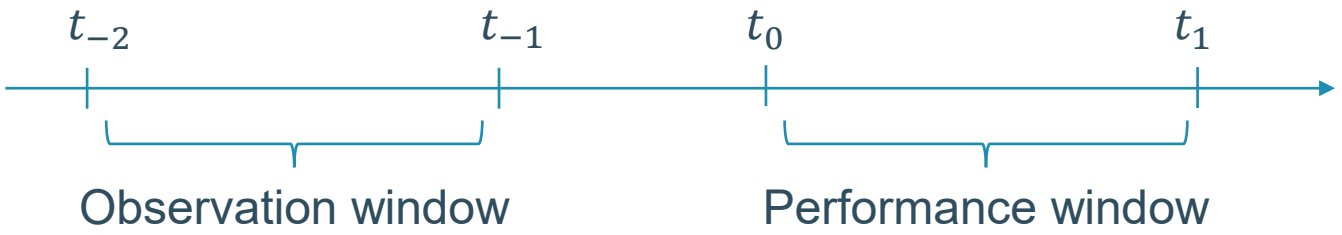
- t_0: first snapshot, time of prediction
- t_1: second snapshot, time of observing outcome (over preceding time window)

Case: customer churn

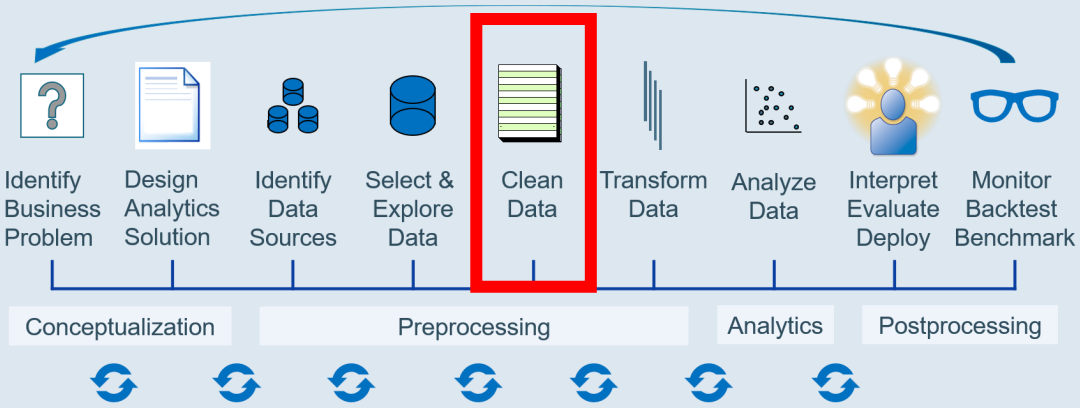


- Better setup?

Design the analytics solution!



- Early warning: reaction time!
- Lost causes versus persuadables (cf. Prescriptive analytics: uplift modeling)



5. Clean data

Missing values

Outliers

Missing values

ID	Age	Income	Housing	Credit score	Default
12580	45	2600	Own	620	Good
42882	?	1100	Own	?	Bad
42440	27	1400	?	?	Good
24580	57	3200	Rent	700	Good
31008	?	1800	Other	?	Good
21007	42	1600	Rent	?	Good
67088	21	1900	?	?	Bad
45880	24	1500	Own	350	Good
…	…	…	…	…	…

Missing values: treatment

1. **Remove**
   - Eliminate rows (case-wise deletion)
   - Eliminate columns

- But!
  - Many missing values?
  - Removal of useful information?
  - Introduction of bias?

Missing values: treatment

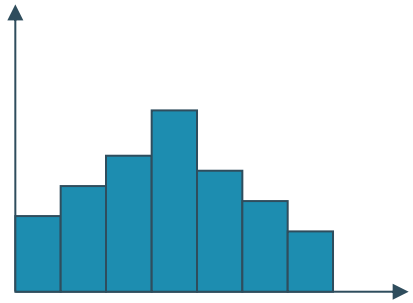
2. **Replace**

- Acquire ‘true’ values
  - Contact customer
  - Purchase external data
  - Note: expensive - effort

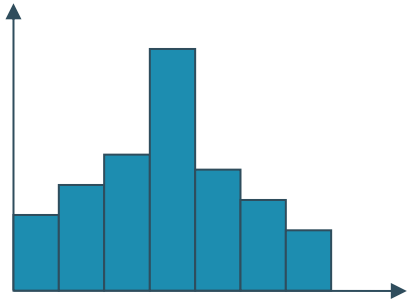
- **Imputation** techniques
  - Based on the non-missing data, missing values can be imputed

Missing values: imputation techniques

- **Impute** with the **mean** or, better, the **median** for continuous variables
- Impute with the **mode** (most frequently occuring value) for categorical variables
- Note: impact on distribution:



Before imputation

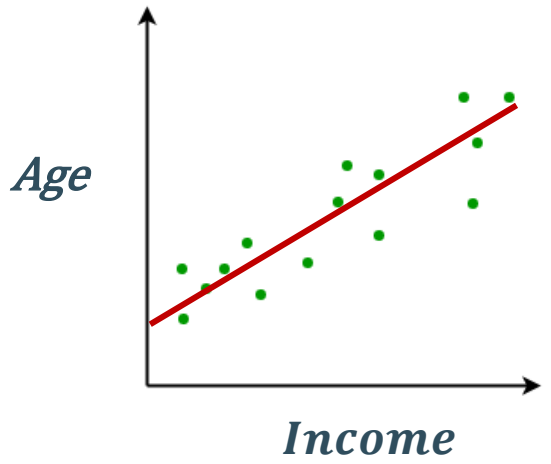


After imputation

Missing values: imputation techniques

- **Impute** by developing a **prediction model**
- E.g.:

ID	Age	Income	Housing	Credit score	Default
12580	45	2600	Own	620	Good
42882	?	1100	Own	?	Bad
42440	27	1400	?	?	Good
24580	57	3200	Rent	700	Good
31008	?	1800	Other	?	Good
21007	42	1600	Rent	?	Good
67088	21	1900	?	?	Bad
45880	24	1500	Own	350	Good
…	…	…	…	…	…



Missing values: imputation techniques

- **Pro:**
  - Allows to apply techniques that require complete data sets
  - No loss of (correct) information
- **Con**:
  - Imputation introduces (additional) errors
  - Is it worth the effort, given the additional complexity?

Note:

Data analytics: pragmatic approach!
vs.
Statistics: rigorous approach!

Best to 'optimally combine' … but … how?

Missing values: imputation techniques

3. **Keep**
   - Categorical variables: new category '*Missing*'
   - Continuous variables:
     - Introduce **dummy variable** 'missing value indicator' (e.g., CS_Miss)
     - + replace

ID	Age	Income	Housing	Credit score	Default
12580	45	2600	Own	620	Good
42882	?	1100	Own	?	Bad
42440	27	1400	?	?	Good
24580	57	3200	Rent	700	Good
31008	?	1800	Other	?	Good
21007	42	1600	Rent	?	Good
67088	21	1900	?	?	Bad
45880	24	1500	Own	350	Good
…	…	…	…	…	…



ID	Age	Age_Miss	Income	Housing	Credit score	CS_Miss	Default
12580	45	0	2600	Own	620	0	Good
42882	**35**	1	1100	Own	**620**	1	Bad
42440	27	0	1400	**Missing**	**620**	1	Good
24580	57	0	3200	Rent	700	0	Good
31008	**35**	1	1800	Other	**620**	1	Good
21007	42	0	1600	Rent	**620**	1	Good
67088	21	0	1900	**Missing**	**620**	1	Bad
45880	24	0	1500	Own	350	0	Good
…	…	…	…	…	…	…	…

Missing values: imputation techniques

4. **Weights-of-evidence** (cf. infra)

The profesor's favorite

Outliers

- Extreme values
  - E.g.:
    - Age = 200
    - Income = 1,000,000€

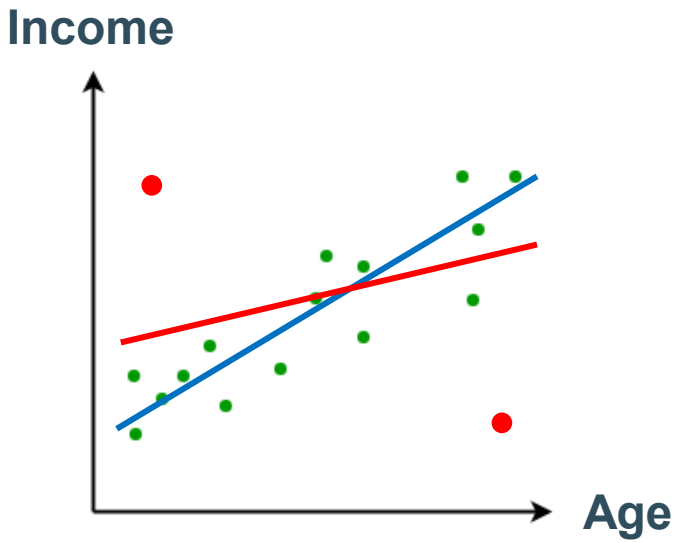
- Outliers can be the result of data entry errors
  - To be treated as missing values

Outliers

Why should we care about outliers?

- Potentially, outliers have a **disproportional impact** on the result
  - Some techniques are **sensitive** to outliers
  - E.g.: OLS regression modeling

- **Detection** versus **treatment**



Outliers: detection

Detecting outliers of categorical variables?

How to *detect*?

- Visual approaches: histograms, boxplots
- z-score:
  - Mean μ
  - Standard deviation σ
  - $z = \frac{x-\mu}{\sigma}$
  - $z =$ # standard deviations from the mean
  - Rule-of-thumb: outlier if $|z| > 3$

The profesor's favorite

	Age	z-score
	45	0,77
	35	-0,06
	27	-0,73
	57	1,77
	35	-0,06
	42	0,52
	21	-1,23
	24	-0,98
	80	3,68
Mean	**35,75**	**0**
St.Dev.	**12,01**	**1**

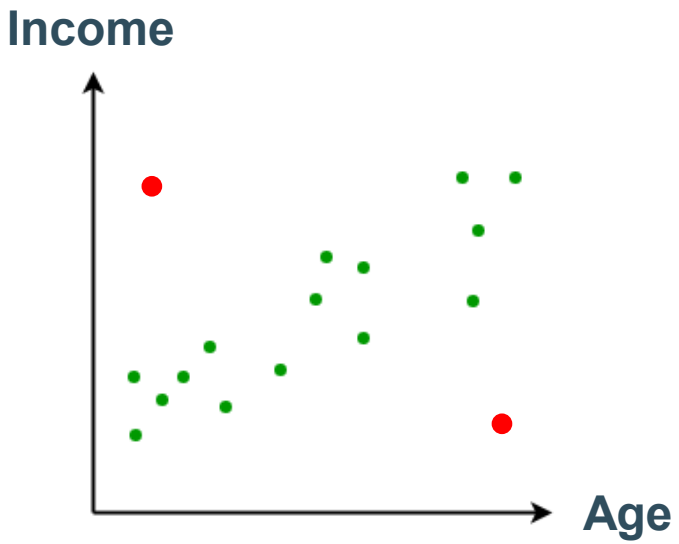
Outliers: treatment

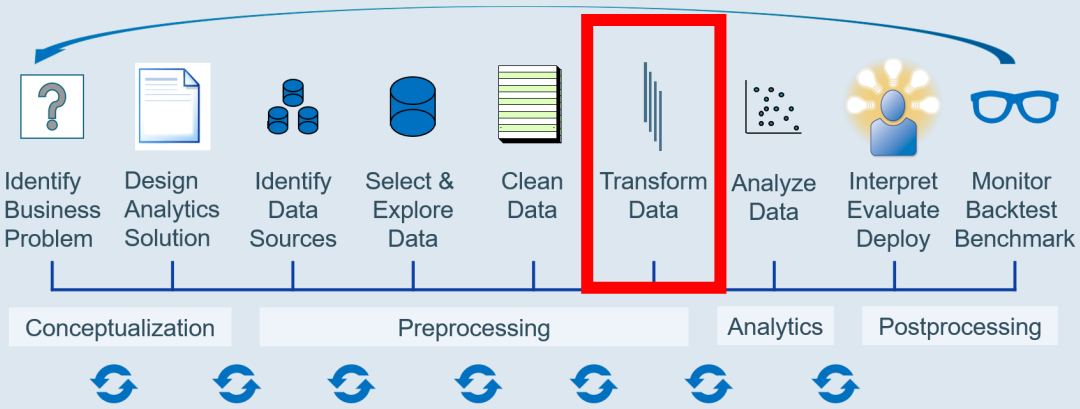
How to *treat*?

- I.e., how to **reduce impact**?
- Make the value less extreme: **truncation** (a.k.a. winsorizing or capping)
- E.g.:
  - If $z > 3$, then replace with value for which $z = 3$
  - Replace with 99th percentile value
  - …
- **Note**: outliers may be predictive for the class of interest!
  - E.g.: fraud detection

Outliers: multivariate outliers

- **Multivariate outliers**
  - Extreme or unusual values, when looking at multiple dimensions simultaneously
  - E.g.: age vs. income
- **Multivariate outlier detection**
  - Anomaly detection approaches
  - Mahalanobis distance
- **Multivariate outlier treatment**
  - Imputation with a 'normal' value?
  - Multivariate regression, autoencoders, …
- **Practical advice: ignore multivariate outliers**





6. Transform data

Featuring engineering

Variable transformation

Transform data

What is meant with '*result*'?

Why should (clean) data be *transformed*?

1. **Improving the *result***
   - To increase predictive performance
   - To improve interpretability
   - …
2. **Technical reasons**:
   - Method sensitive to the **scale of variables**
     → Make feature ranges comparable (e.g., normalization)
   - Transform **categorical** variables into **continuous** variables
   - Transform **continuous** variables into **categorical** variables
   - …

Feature engineering

Feature engineering

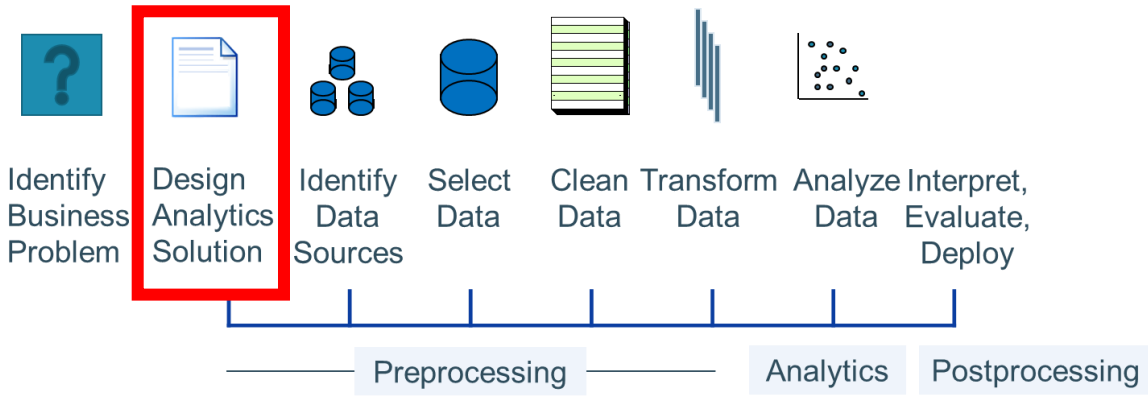
Why feature engineering?

- Enrich data set so as to increase predictive performance
- **Exercise**: credit card transactional fraud detection

Customer ID	Merchant ID	Merchant Type	Time stamp	Date	Amount	Channel type	Terminal ID
005489	AC45654	D01	18:24:19	22:02:19	234.45	A	ZT01
004657	XS45641	C02	54:52:12	22:02:19	17.50	A	CT20
002545	SD12155	C08	02:04:11	22:0:19	54.99	C	CR12

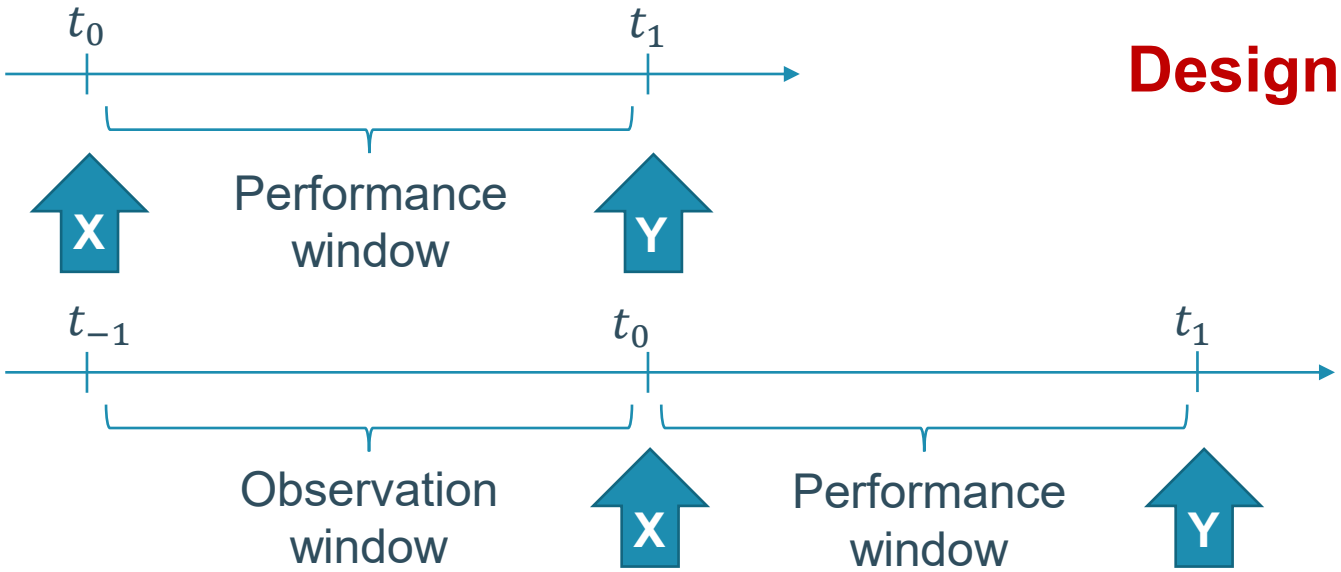
- To improve interpretability and actionable insight
  - Cf. infra: to explain, to support remedial actions
  - E.g.: why do customers churn? Network quality? Social influence?

Case: customer churn



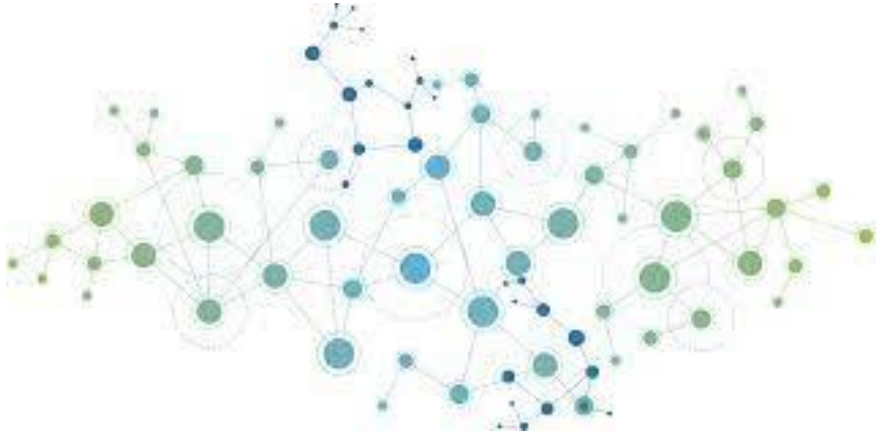
Design the analytics solution!

- Featurization: time dimension



- **Time-flattening** the data / the problem: **removing the time dimension** by **defining features** that **summarize performance period**
  - Recency – Frequency – Monetary (RFM) variables

Feature engineering



- Transforming **unstructured** to **tabular** data
  - Structured or network data (network analytics)
  - Text data
  - Images
  - …
- Representation learning: cf. Chapter 6: Artificial Neural Networks
  - Embeddings
- **Why?**
  - E.g., predictive power
  - Technical reason: integrating unstructured data (~categorical > continuous)

PRAGMA: Revolut Foundation Model

Maxim Ostroukhov[1] **Ruslan Mikhailov**[1] **Vladimir Iashin**[1] **Artem Sokolov**[1]
Andrei Akshonov[1] **Vitaly Protasov**[1] **Andrey Goncharov**[1] **Dmitrii Beloborodov**[1]
Vince Mullin[2] **Roman Y. Enzmann**[2] **Georgios Kolovos**[2] **Jason Renders**[2]
Pavel Nesterov[1] **Anton Repushko**[1]

[1]*Revolut Research* [2]*NVIDIA*

Modern financial systems generate vast quantities of transactional and event-level data that encode rich economic signals. This paper presents PRAGMA, a family of foundation models for banking event sequences. Our approach pre-trains a Transformer-based architecture with masked modelling on a large-scale, heterogeneous banking event corpus using a self-supervised objective tailored to the discrete, variable-length nature of financial records. The resulting model supports a wide range of downstream tasks such as credit scoring, fraud detection, and lifetime value prediction: strong performance can be achieved by training a simple linear model on top of the extracted embeddings and can be further improved with lightweight fine-tuning. Through extensive evaluation on downstream tasks, we demonstrate that PRAGMA achieves superior performance across multiple domains directly from raw event sequences, providing a general-purpose representation layer for financial applications.

Disclaimer: We report only relative improvements, as absolute metrics are commercially sensitive.
All examples are synthetic and not from real production data.

PRAGMA: Revolut Foundation Model

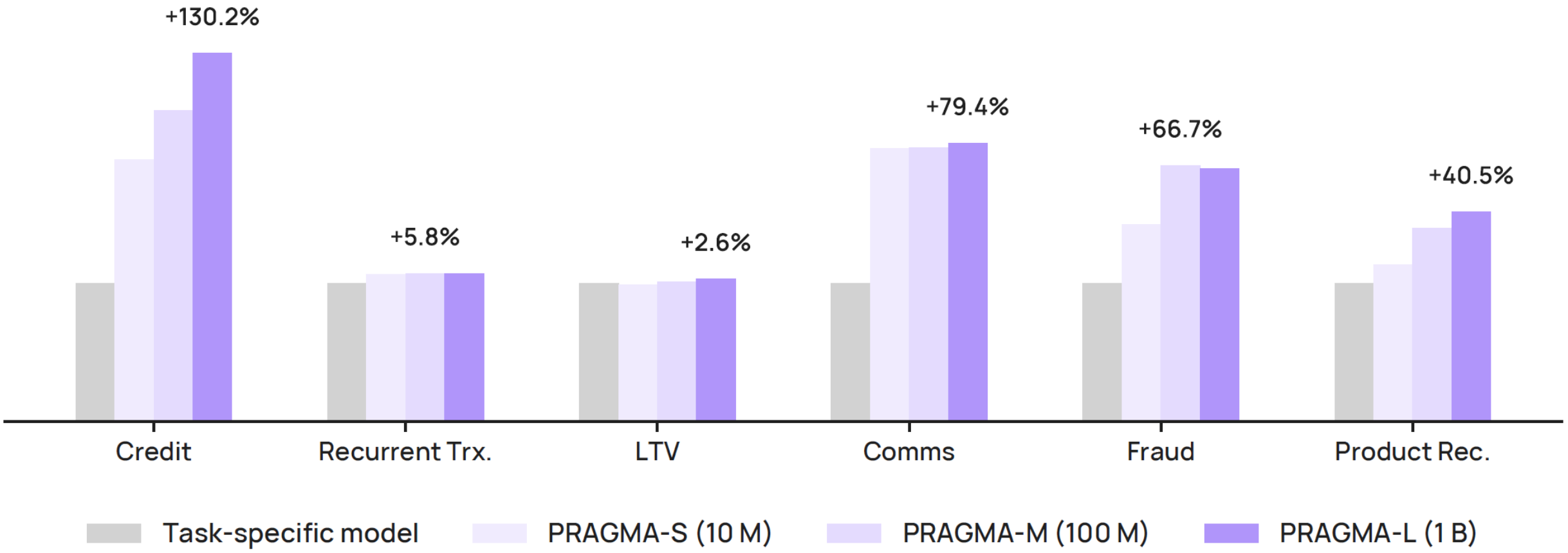


Figure 1: A single architecture from 10M to 1B parameters that outperforms task-specific models across tasks.

Variable transformation: normalization

- **Normalization?**
  - Data processing technique
    - To **rescale the values of a variable** to a **common range** or
    - To **transform its distribution** to be more closely aligned with a **Gaussian distribution**

- **Why**?
  - Some techniques are **sensitive to the scale of variables**
  - Resulting in variables to be more important only because of their scale!
    - E.g.: ANNs: normalizing the data improves the training process (cf. Chapter 6)
    - E.g.: clustering using Euclidean distance (cf. Chapter 8)

https://sebastianraschka.com/Articles/2014_about_feature_scaling.html
https://towardsdatascience.com/normalization-vs-standardization-quantitative-analysis-a91e8a79cebf

Variable transformation: normalization

- **Normalization?**
  - Data processing technique
    - To **rescale the values of a variable** to a **common range**

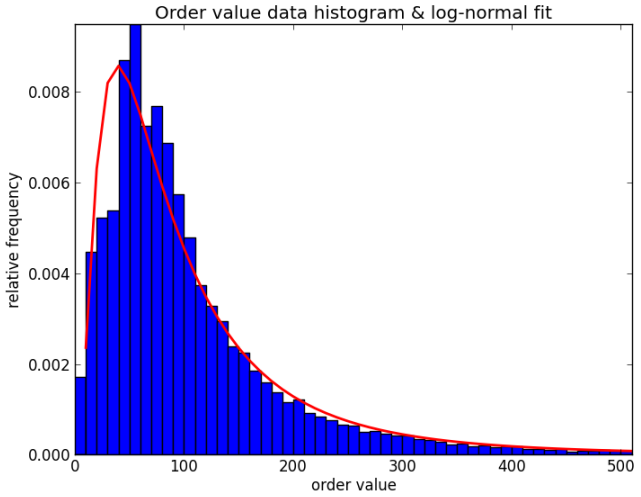
- E.g., **min-max normalization**: x → [min,max]
  - Rescale variables (e.g., to range [0,1]) with transformation: $x_{norm} = \frac{(x - x_{min})}{(x_{max} - x_{min})}$
  - Example:
    - $x = \{150, 250, 50, 100, 150, 550\}$
    - $x_{min} = 50$
    - $x_{max} = 550$
    - $x_{norm} = \{0.2, 0.4, 0, 0.1, 0.2, 1\}$

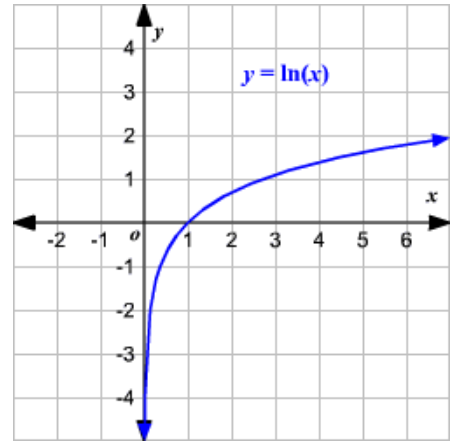
https://sebastianraschka.com/Articles/2014_about_feature_scaling.html
https://towardsdatascience.com/normalization-vs-standardization-quantitative-analysis-a91e8a79cebf

Variable transformation: normalization

- **Normalization?**
  - Data processing technique
    - To **rescale the values of a variable** to a **common range**
    - To **transform its distribution** to be more closely aligned with a **Gaussian distribution**

- E.g., **logarithmic transformation**
  - To **reduce right skewness** of data
  - To "facilitate", e.g., linear regression
    - Better model fit (nonlinear > linear relation)
    - Less impact of outliers
    - Normality of errors for significance testing





Variable transformation: standardization

	Age	z-score
	45	0,77
	35	-0,06
	27	-0,73
	57	1,77
	35	-0,06
	42	0,52
	21	-1,23
	24	-0,98
	80	3,68
Mean	**35,75**	**0**
St.Dev.	**12,01**	**1**

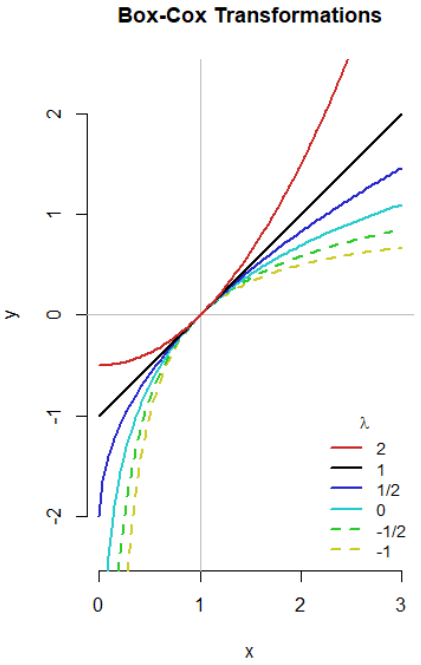
- **Standardization**?
  - Rescale data to have a **mean of zero** and a **standard deviation of one**
  - Values measure how many standard deviations the value is from its mean
  - Creating a common scale for comparison and analysis
  - = z-score standardization

- **Standardization or normalization**?
  - Depends on the application
  - Impact of outliers on normalization is different from standardization

https://sebastianraschka.com/Articles/2014_about_feature_scaling.html
https://towardsdatascience.com/normalization-vs-standardization-quantitative-analysis-a91e8a79cebf

Variable transformation

- **Advanced transformations:**
  - Box-Cox
  - Yeo-Johnson
  - Principal Component Analysis
  - …
- **Note:**
  - Increased ***complexity***
  - Reduced ***interpretability*** of the resulting model
  - Improved ***performance*** **???**



$$y_i^{(\lambda)} = \begin{cases} \dfrac{y_i^{\lambda} - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \ln(y_i) & \text{if } \lambda = 0, \end{cases}$$

Cox, N. J.: Transformations: an introduction - http://fmwww.bc.edu/repec/bocode/t/transint.html

Variable transformation: categorical → continuous

- **Categorical variable**: e.g., *Housing* variable
  - How to include in a linear or logistic regression model?
  - **Categorical to continuous variable transformation!**
  - How?
    - ~~Numerical encoding~~
    - **Dummy encoding**
    - **Target encoding**
  - Numerical encoding:
    - Own = 1, Rent = 2, Other = 3, …

ID	Age	Income	Housing	Credit score	Default
12580	45	2600	Own	620	Good
42882	?	1100	Own	?	Bad
42440	27	1400	Own	?	Good
24580	57	3200	Rent	700	Good
31008	?	1800	Other	?	Good
21007	42	1600	Rent	?	Good
67088	21	1900	Own	?	Bad
45880	24	1500	Own	350	Good
…	…	…	…	…	…

Variable transformation: dummy encoding

- **Dummy encoding**
  - A.k.a. one-hot encoding

Collinearity!
Instability!

Housing	Housing_own	Housing_rent	Ho…ing_…ther
Own	1	0	
Rent	0	1	
Other	0	0	1

ID	Age	Income	Housing_own	Housing_rent	Credit score	Default
12580	45	2600	1	0	620	Good
42882	?	1100	1	0	?	Bad
42440	27	1400	1	0	?	Good
24580	57	3200	0	1	700	Good
31008	?	1800	0	0	?	Good
21007	42	1600	0	1	?	Good
67088	21	1900	1	0	?	Bad
45880	24	1500	1	0	350	Good
…	…	…	…	…	…	…

- **Stability or variance**
  - Level of perturbation of a **machine learning algorithm** by small changes to its inputs
  - A **stable** learning algorithm is desirable: resulting model and predictions do not change much when the training data is modified slightly (vs. unstable)
  - ~ overfitting & generalization power (cf. Chapters 3 & 5)
- **Multicollinearity → Instability!**
  - The **dummy variable trap**
  - **n** categories → **n-1** dummies!
- **Many dummies → Instability!**
  - Reduce number of dummies?

https://machinelearningmastery.com/why-one-hot-encode-data-in-machine-learning/
https://en.wikipedia.org/wiki/Stability_(learning_theory)

Variable transformation: dummy encoding



Variable transformation: dummy encoding

- **To reduce instability** we can:
  - Reduce the number of dummies
    - By reducing the number of categories of categorical variables
    - By reducing the number of categorical variables
  - Use **target encoding** instead of dummy encoding

- **How to reduce the number of categories of a categorical variable?**
  - Coarse classification using pivot tables
  - Coarse classification using chi-squared analysis

- **How to reduce the number of categorical variables?**
  - Variable selection, e.g., using a filter such as the information value

Coarse classification: pivot tables

- Coarse classification using pivot tables > odds > eye-balling

Data table

ID	Age	Income	Housing	Credit score	Default
12580	45	2600	Own	620	Good
42882	?	1100	Own	?	Bad
42440	27	1400	Own	?	Good
24580	57	3200	Rent	700	Good
31008	?	1800	Other	?	Good
21007	42	1600	Rent	?	Good
67088	21	1900	Own	?	Bad
45880	24	1500	Own	350	Good
…	…	…	…	…	…

Which regrouping is the best?

Pivot table

Housing:	Own	Rent unfurnished	Rent furnished	With parents	Other	No answer	Total
Class = Good	6000	1600	350	950	90	10	9000
Class = Bad	300	400	100	100	50	10	1000
Good:Bad odds	**20:1**	**4:1**	**2.5:1**	**9.5:1**	**1.8:1**	**1:1**	**9:1**

Example adopted from *Credit Scoring and Its Applications*, by Thomas, Edelman and Crook, 2002

Coarse classification: chi-squared test statistic

- Suppose we want to regroup in three categories, which option is best?
  - **Option 1**:
    - Owners (own)
    - Renters (rent unfurnished and furnished),
    - Others (with parents, other, no answer)
  - **Option 2**:
    - Owners (own)
    - With parents (with parents)
    - Others (rent unfurnished, rent furnished, other, no answer)

Housing:	Own	Rent unfurnished	Rent furnished	With parents	Other	No answer	Total
Class = Good	6000	1600	350	950	90	10	9000
Class = Bad	300	400	100	100	50	10	1000
Good:Bad odds	**20:1**	**4:1**	**2.5:1**	**9.5:1**	**1.8:1**	**1:1**	**9:1**

Coarse classification: chi-squared test statistic

Option 1:

- A first table shows the **empirical frequencies**:
- The **independence frequencies** can be calculated from the empirical frequencies as follows:
  - E.g., the **expected** number of **good owners** if the odds were the same for owners as for the whole population:

    6300 * 9000/10000 = 5670
  - Calculating the expected number for each cell gives the independence frequencies table:
- The **chi-squared distance** between both tables:

Empirical frequencies

Housing:	Own	Renters	Others	Total
Class = Good	6000	1950	1050	9000
Class = Bad	300	540	160	1000
Total	**6300**	**2490**	**1210**	**10000**



Independence frequencies

Housing:	Own	Renters	Others	Total
Class = Good	5670	2241	1089	9000
Class = Bad	630	249	121	1000
Total	**6300**	**2490**	**1210**	**10000**

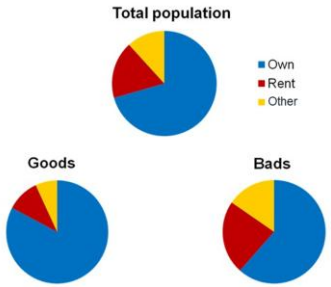


$$\chi^2 = \frac{(6000-5670)^2}{5670} + \frac{(300-630)^2}{630} + \frac{(1950-2241)^2}{2241} + \frac{(540-249)^2}{249} + \frac{(1050-1089)^2}{1089} + \frac{(160-121)^2}{121} = 583$$

Coarse classification: chi-squared test statistic

- Likewise, **option 2** gives this:

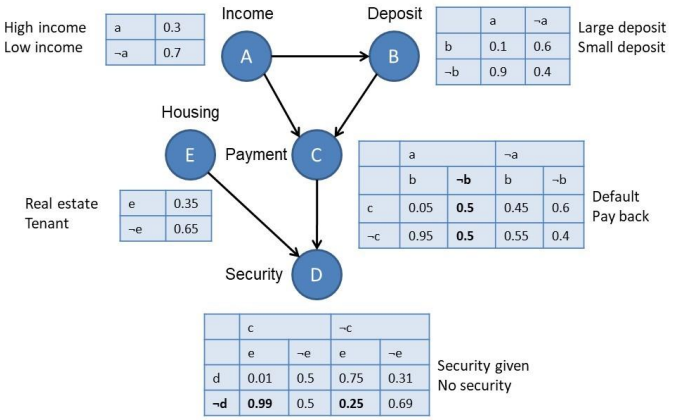
$$\chi^2 = \frac{(6000-5670)^2}{5670} + \frac{(300-630)^2}{630} + \frac{(950-945)^2}{945} + \frac{(100-105)^2}{105} + \frac{(2050-2385)^2}{2385} + \frac{(600-265)^2}{265} = 662$$



- The **higher the test statistic, the better the split!**
  - The higher the statistic, the **more different the distribution of the goods and the bads** across the categories
  - Best predictor? Full separation between distribution of goods and bads!
    - E.g., all own = good, all rent & other = bad
- **Conclusion**: option 2 is the better split!
- Note: formal test possible by comparing with chi-square distribution with k-1 degrees of freedom for k classes of the characteristic

Variable transformation: categorical ← continuous

- **Continuous to categorical: why?**
  - Techniques which require the variables to be categorical
    - E.g.: Bayesian network
  - Interpretability of model or business requirements
    - E.g.: Business users prefer age segments
    - E.g.: Need to transform final model into scorecard (cf. Part 2: credit risk modeling)



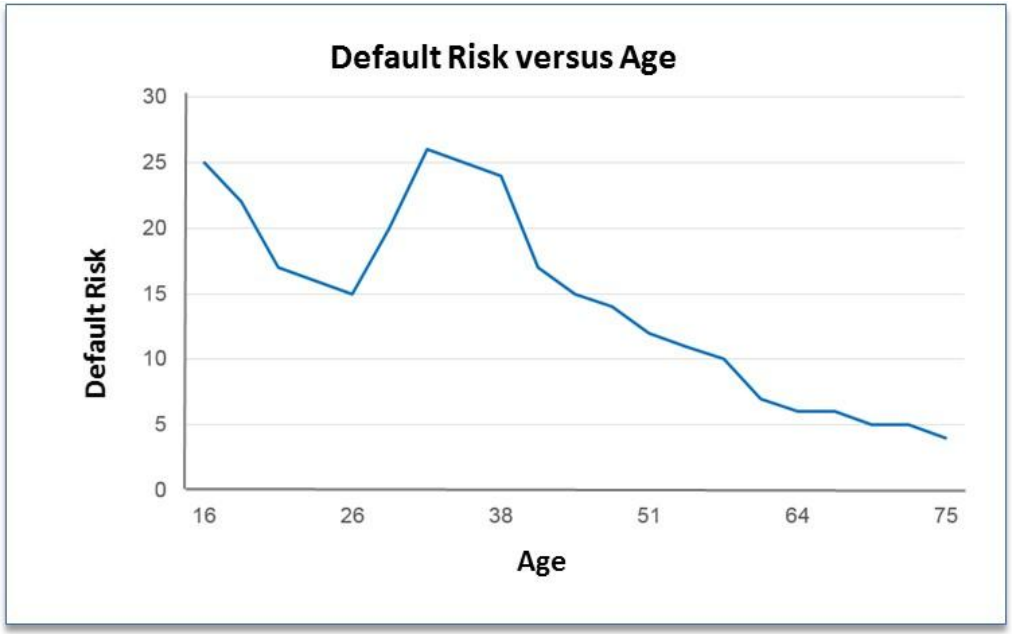
Example Scorecard Criteria	Range	Points
Business Stability	0-1 Year	0
	0-2 Years	10
	0-3 Years	20
	3+ Years	25
Bureau Score	0	0
	600-650	0
	700-750	30
	750+	50
Bank Account Stability	0-1 Year	0
	0-2 Years	10
	0-3 Years	20
	3+ Years	25
Total Score	XXX	

Variable transformation: categorical ← continuous

- **Continuous to categorical: why?**
  - For part of the observations only category is known (e.g., *older* observations collected with a previous version of a software, external data, …)
    - E.g.: age category
  - Anonymization (GDPR)
    - E.g.: income class instead of income (€1000 - €1500, €1501 - €2000, …)

Variable transformation: categorical ← continuous

- **Continuous to categorical: why?**
  - Incorporate **non-linear relations within** a **linear model** to improve model performance while **maintaining interpretability**



Q: then why don't you 'simply' fit a nonlinear model?

Thomas, L. C. (2000). A survey of credit and behavioural scoring: forecasting financial risk of lending to consumers. International journal of forecasting, 16(2), 149-172.

Variable transformation: categorical ← continuous

- **How?**
  - **Binning:**
    - Equal width
    - Equal interval
  - **But:**
    - Binning: does not take the objective into account!

- **Note: continuous → categorical …**
  - Introduces **need for more dummy variables**
    - If technique cannot handle categorical variables
  - Increases instability!

- **Smarter approach?**
  - **Target encoding** with Weights-of-evidence + Information Value-driven binning

Target encoding: weights-of-evidence

$$\text{ODDS_cat} = \left(\frac{Goods_cat}{Bad_cat}\right)$$

Age_cat	Count	Distr. Count	Goods	Distr. Goods	Bads	Distr. Bads	WOE
Missing	240	2,18%	200	2,00%	40	4,00%	-69,31
18-22	2400	21,82%	2000	20,00%	400	40,00%	-69,31
23-26	2475	22,50%	2183	21,83%	292	29,20%	-29,09
27-29	424	3,85%	368	3,68%	56	5,60%	-41,99
30-35	2659	24,17%	2530	25,30%	129	12,90%	67,36
35-44	1864	16,95%	1807	18,07%	57	5,70%	115,38
44+	938	8,53%	912	9,12%	26	2,60%	125,50
Total:	**11000**		**10000**		**1000**		

Outliers?

$$\text{WOE} = \ln\left(\frac{Distr.Good}{Distr.Bad}\right) \times 100$$

Why do we take the logarithm of the "relative odds" and not of the "absolute odds"?

→ **Then the WOE values that are obtained are independent of the class distribution!**

→ **Then the WOE values permit easy interpretation – negative versus positive values**

Target encoding: weights-of-evidence

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE		Odds
Owner	6300	63%	6000	67%	300	30%	79,85		20
Rent unfurnished	2000	20%	1600	18%	400	40%	-81,09		4
Rent furnished	490	5%	350	4%	140	14%	-128,09		2,5
With parents	1050	11%	950	11%	100	10%	5,41		9,5
Other	140	1%	90	1%	50	5%	-160,94		1,8
No answer	20	0%	10	0%	10	1%	-219,72		1
Total	10000		9000		1000				

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE		Odds
Owner	6600	66%	6000	67%	600	30%	79,85		10
Rent unfurnished	2400	24%	1600	18%	800	40%	-81,09		2
Rent furnished	630	6%	350	4%	280	14%	-128,09		1,25
With parents	1150	12%	950	11%	200	10%	5,41		4,75
Other	190	2%	90	1%	100	5%	-160,94		0,9
No answer	30	0%	10	0%	20	1%	-219,72		0,5
Total	11000		9000		2000				

Why do we take the logarithm of the "relative odds" and not of the "absolute odds"?

→ Then the WOE values that are obtained are independent of the class distribution!

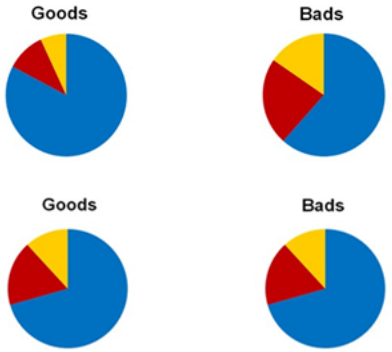
→ Then the WOE values permit easy interpretation – negative versus positive values

Target encoding: weights-of-evidence

- What if the **proportion of the goods** = **proportion of bads** in a category?
  - Then the relative odds = 1 and WOE = 0
- What if the **proportion of the goods** = **proportion of bads** in **each** category?
  - Then WOE = 0 in each category
  - No predictive power in variable!
    - Belonging to a category does not reveal any information on the class

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE
Owner	6300	63%	5.670	63%	630	63%	0,00
Rent unfurnished	2000	20%	1.800	20%	200	20%	0,00
Rent furnished	490	5%	441	5%	49	5%	0,00
With parents	1050	11%	945	11%	105	11%	0,00
Other	140	1%	126	1%	14	1%	0,00
No answer	20	0%	18	0%	2	0%	0,00
Total	10000		9000		1000		

Target encoding: weights-of-evidence



- Measure of **predictive power**?
- **Information Value (IV):**

$$IV = \sum_{category} \left((\text{Distr. Good}_{category} - \text{Distr. Bad}_{category}) * \text{woe}_{category} / 100 \right)$$

Absolute difference in conditional probability of age bin, conditional on good/bad

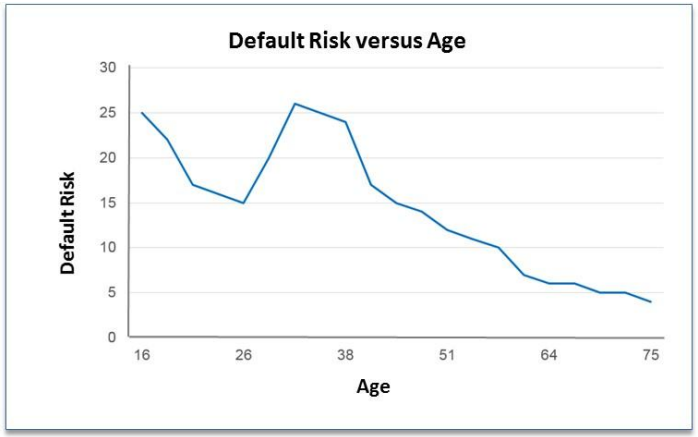
Relative difference in conditional probability of age bin, conditional on good/bad ($woe_{cat}/100 = \ln(ratio)$)

- Rule of thumb:
  - IV < 0.02 unpredictive
  - 0.02 < IV < 0.1 weak
  - 0.1 < IV < 0.3 medium
  - IV > 0.3 strong

Age	Count	Distr. Count	Goods	Distr. Goods	Bads	Distr. Bads	WOE	IV
Missing	240	2,18%	200	2,00%	40	4,00%	-69,31	0,0139
18-22	2400	21,82%	2000	20,00%	400	40,00%	-69,31	0,1386
23-26	2475	22,50%	2183	21,83%	292	29,20%	-29,09	0,0214
27-29	424	3,85%	368	3,68%	56	5,60%	-41,99	0,0081
30-35	2659	24,17%	2530	25,30%	129	12,90%	67,36	0,0835
35-44	1864	16,95%	1807	18,07%	57	5,70%	115,38	0,1427
44+	938	8,53%	912	9,12%	26	2,60%	125,50	0,0818
Total:	**11000**		**10000**		**1000**		**IV:**	**0,49**

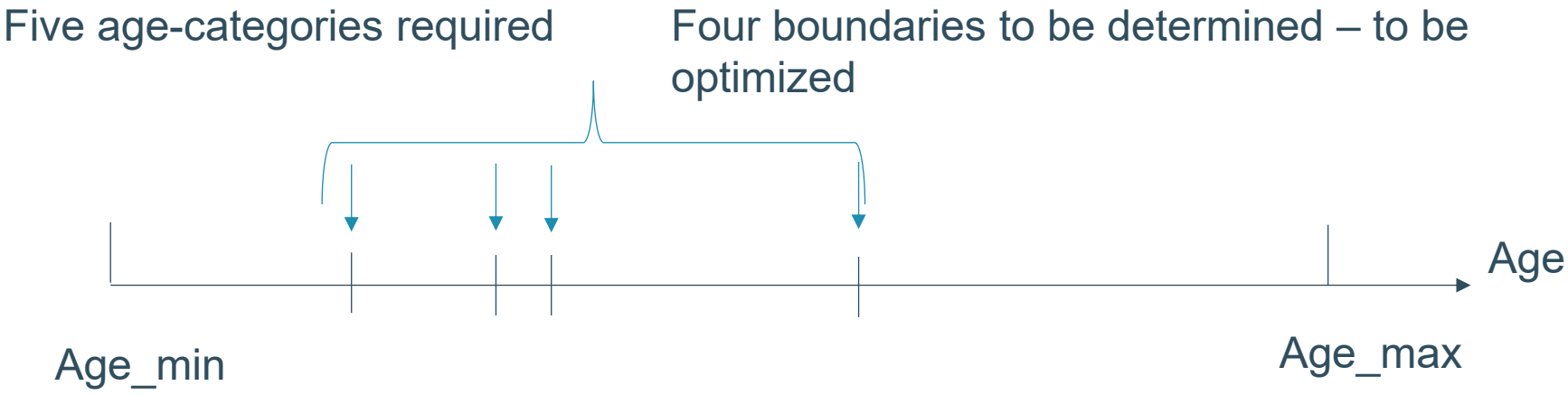
x10 x10

Target encoding: weights-of-evidenc



Age → Age_cat ?

- Continuous → Categorical: **category-boundaries**?
  - For a **given number of categories**,
  - the boundaries can be optimized,
  - so as to **maximize** the **predictive power**,
  - in terms of **IV** (→ IV-driven binning)



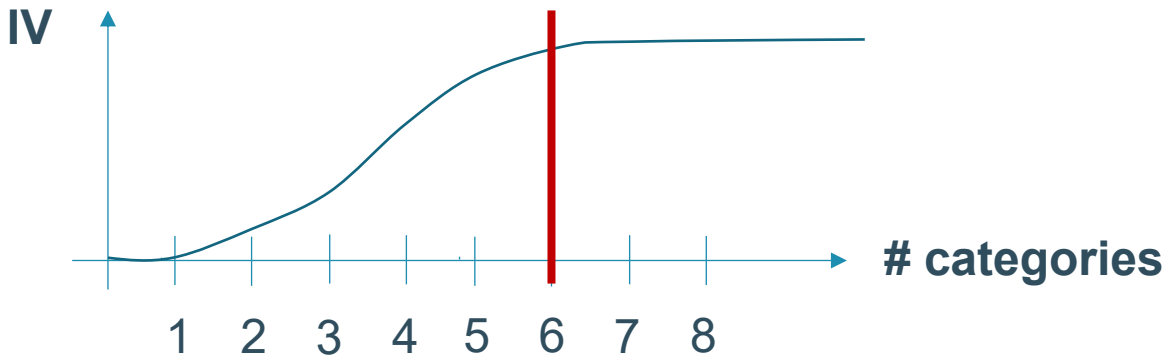
Target encoding: weights-of-evidence

- **Number of categories?**
  - Trade-off:
    - Fewer categories preferred because of **interpretability** and **stability**
    - More categories to keep as much **predictive power** as possible
      - Note: When binning to a single category, all predictive power is lost
      - For an increasing number of categories, a gain in predictive power is achieved
      - Up until some point, when further increasing the number of categories does not add **significant** predictive power

Extreme case reasoning → to gain intuitive understanding of how approaches work

Target encoding: weights-of-evidence

- **Number of categories?**
  - Practical solution: perform sensitivity analysis
    - IV versus number of categories
    - Decide on cut-off: elbow point?



Target encoding: weights-of-evidence

- **Number of categories?**
  - Trade-off:
    - Fewer categories preferred because of **interpretability** and **stability**
    - More categories to keep as much **predictive power** as possible
      - Note: When binning to a single category, all predictive power is lost
      - For an increasing number of categories, a gain in predictive power is achieved
      - Up until some point, when further increasing the number of categories does not add **significant** predictive power

Stability of WOE?
- **The fewer values in a category**
- **The less reliable/robust/stable the WOE value that is calculated**

Target encoding: weights-of-evidence

- Note: **stability** of WOE values!

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE
Owner	6300	63%	6000	67%	300	30%	79,05
Rent unfurnished	2000	20%	1600	18%	400	40%	-81,90
Rent furnished	490	5%	350	4%	140	14%	-128,90
With parents	1050	11%	950	11%	100	10%	4,60
Other	140	1%	90	1%	50	5%	-161,75
No answer	2	0%	1	0%	1	0%	-220,53
Total	9982		8991		991		

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE
Owner	6301	63%	6001	67%	300	30%	79,06
Rent unfurnished	2000	20%	1600	18%	400	40%	-81,90
Rent furnished	490	5%	350	4%	140	14%	-128,90
With parents	1050	11%	950	11%	100	10%	4,60
Other	140	1%	90	1%	50	5%	-161,75
No answer	3	0%	2	0%	1	0%	-151,21
Total	9984		8993		991		

One extra observation leads to significant change in WOE value!

If your “model” changes when the data changes, then your model is unstable (cf. supra) – which we do not want, as the model is to capture relations that are independent of the data sample at hand and describe the population!

Target encoding: weights-of-evidence

- Increasing **stability** with **Laplace smoothing**:

$$WOE = \ln\left[\frac{\text{Distr Good}}{\text{Distr Bad}}\right] \times 100$$

$$WOE = \ln\left[\frac{\text{Distr Good} + n}{\text{Distr Bad} + n}\right] \times 100$$

Simpler solution: require bins with at least 5% of the observations (cf. Demo)

- n: smoothing parameter
- Larger n?
  - → More smoothing, less reliance on data in calculating WOE
  - → WOE closer to zero, meaning no increased or decreased risk (a-priori risk)
  - →There needs to be more evidence in the data, for getting to higher WOE values

Moeyersoms, J. (2016). *The value of customer behavior data in predictive modeling*. Universiteit Antwerpen (Belgium).
E.Zdravevski, P. Lameski, A.Kulakov,Weight of evidence as a tool for attribute transformation in the preprocessing stage of supervised learning algorithms, The 2011 International Joint Conference on Neural Networks (IJCNN), 2011, pp. 181–188.

Target encoding: weights-of-evidence

- What value of **n** to use? Do we get a better model? Better predictions?

Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE	WOE_smooth (n = 0,1)
Owner	6300	63%	6000	67%	300	30%	79,05	64,47
Rent unfurnished	2000	20%	1600	18%	400	40%	-81,90	-59,44
Rent furnished	490	5%	350	4%	140	14%	-128,90	-55,20
With parents	1050	11%	950	11%	100	10%	4,60	2,34
Other	140	1%	90	1%	50	5%	-161,75	-31,31
No answer	2	0%	1	0%	1	0%	-220,53	-0,89
Total	9982		8991		991			

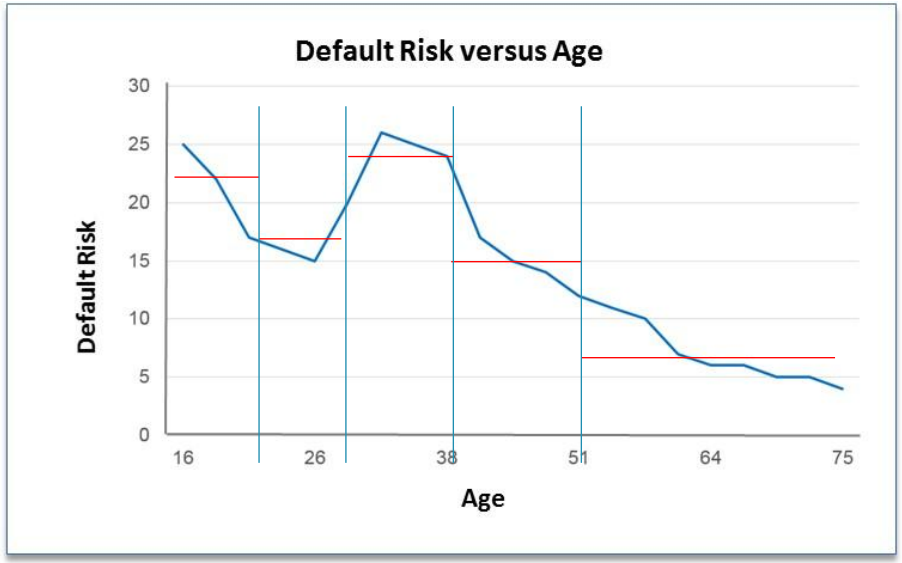
Category	Count	Distr. Count	Good	Distr. Good	Bad	Distr. Bad	WOE	WOE_smooth (n = 0,1)
Owner	6301	63%	6001	67%	300	30%	79,06	64,48
Rent unfurnished	2000	20%	1600	18%	400	40%	-81,90	-59,44
Rent furnished	490	5%	350	4%	140	14%	-128,90	-55,20
With parents	1050	11%	950	11%	100	10%	4,60	2,34
Other	140	1%	90	1%	50	5%	-161,75	-31,31
No answer	3	0%	2	0%	1	0%	-151,21	-0,78
Total	9984		8993		991			

Alternative to smoothing: use fewer bins!!

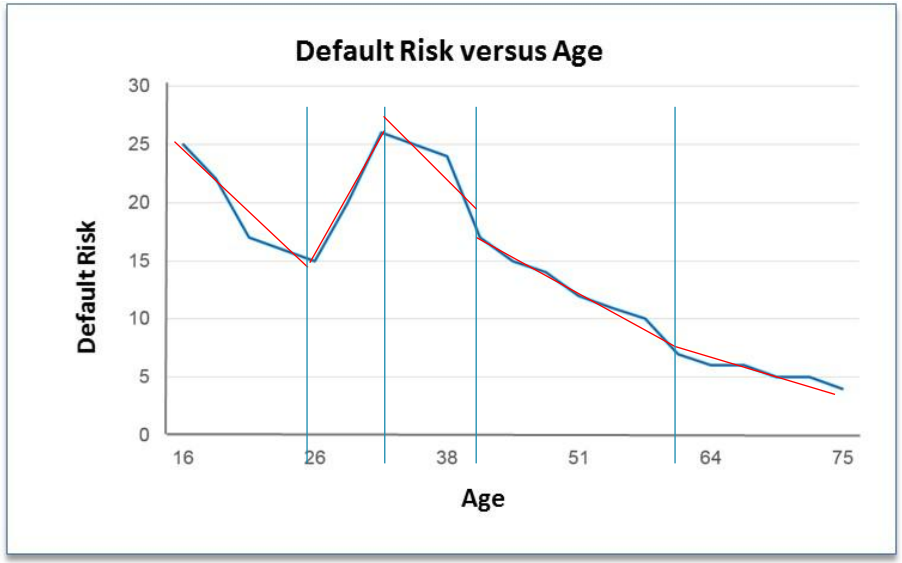
E.g., combine ‘No answer’ and ‘Other’

Target encoding: weights-of-e

- WOE-binning



- With interaction:
  - WOE x age
    (or simply, cat_dummy x age?)
  - Loss of interpretability?



Target encoding: weights-of-evidence

- **WOE: all-in-one solution!**
  - Categorical → continuous
  - Continuous → categorical
  - Missing values
  - Outliers
  - Assessment of predictive strength – variable filter – with IV

The profesor's favorite!

- **Drawbacks?**
  - Some loss of predictive power (? depends – may increase for linear model!)
    → Advanced WOE approach (see, e.g., Raymaekers et al., 2022)

Raymaekers, J., Verbeke, W., & Verdonck, T. (2022). Weight-of-evidence through shrinkage and spline binning for interpretable nonlinear classification. Applied Soft Computing, 115, 108160.

Target encoding for non-binary targets

- WOE: only for binary target!

- **When target is continuous?**
  - Dummy encoding
  - Target encoding?
    - Replace categorical labels with the **average value of the target in the category**

- **When target is multiclass?**
  - E.g.: next word prediction
  - Dummy encoding of the target → multiple outputs → WOE per output ?
    - Multi-task learning (cf. Chapter 6: ANN)

Raymaekers, J., Verbeke, W., & Verdonck, T. (2022). Weight-of-evidence through shrinkage and spline binning for interpretable nonlinear classification. Applied Soft Computing, 115, 108160.

Final remarks

- Transforming variables is feature engineering
- Transforming data is (to date) key in learning from tabular data
- Transforming data is no longer sota in computer vision and NLP
  - Transformers
  - Foundation models
- **Decision trees** incorporate mechanisms for cat. → cont., for variable selection, for handling missing values, outliers, …
  - Cf. Chapter 3!

Final remarks

- Every application context is different,
- Requiring some level of adaptation,
- Requiring a data scientist to have a toolbox of powerful methods,
- Requiring the data scientist to understand the business requirements,
- And requiring the data scientist to align the model with its use

- **Requires to have conceptual understanding and to be creative rather than being able to apply simple “cooking recipes”**

Demo – Customer churn prediction pipeline

- Extended pipeline including:
  - Variable selection in preprocessing step
  - Variable selection with logistic regression: forward, backward, stepwise
- Homework - things to try:
  - Test different approaches towards imputation and variable selection (correlation)
  - Implement outlier detection
  - Implement WOE in the churn prediction pipeline
  - Test ***stability*** of the model: impact of different train/test split on
    - Variable selection
    - Model parameters
    - Model performance

Demo – WOE and IV for credit scoring

- German credit scoring data set

Q&A